rose-leave-disease-6

March 1, 2026

1 Rose Leaf Disease Detection

End-to-end deep learning pipeline for rose leaf disease classification. Architecture: **MobileNetV2** backbone using fixed, reproducible inline hyperparameters.

Outline 1. Data Loading 2. Class Distribution Visualization 3. Clean tf.data Pipeline + Class Weights 4. Model Architecture — MobileNetV2 5. Load Fixed Hyperparameters (Inline, No AutoML) 6. Phase 1 — Classifier Head Training 7. Phase 2 — Full Fine-tuning 8. Training History 9. Model Evaluation 10. Confusion Matrix 11. ROC Curves 12. Per-Class Accuracy 13. Bias & Overfitting Validation 14. Sample Predictions Grid 15. Disease Region Detection & Grad-CAM 16. Save Model 17. TFLite Verification 18. Bias-Reduction Recommendations

[1]:
```
# No extra package installation required.
```

[2]:
```
import os
import random
import numpy as np
import cv2
import tensorflow as tf
import matplotlib.pyplot as plt
import seaborn as sns
from itertools import cycle

from sklearn.model_selection import train_test_split
from sklearn.utils.class_weight import compute_class_weight
from sklearn.metrics import (
    classification_report, confusion_matrix, roc_curve, auc
)
from sklearn.preprocessing import LabelEncoder, label_binarize

#   Reproducibility
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
tf.random.set_seed(SEED)

#   Mixed-precision training (speeds up training on GPU with Tensor Cores)
tf.keras.mixed_precision.set_global_policy('mixed_float16')
```

```python
print('Compute dtype:', tf.keras.mixed_precision.global_policy().compute_dtype)

#   Publication-quality matplotlib settings
plt.rcParams.update({
    'font.family':        'serif',
    'font.size':          12,
    'axes.titlesize':     14,
    'axes.labelsize':     12,
    'xtick.labelsize':    10,
    'ytick.labelsize':    10,
    'legend.fontsize':    10,
    'figure.dpi':         300,
    'figure.facecolor':   'white',
    'axes.facecolor':     'white',
    'axes.grid':          True,
    'grid.alpha':         0.3,
    'savefig.dpi':        300,
    'savefig.bbox':       'tight',
})

print('TensorFlow:', tf.__version__)
```

```
2026-03-01 14:29:22.403184: E
external/local_xla/xla/stream_executor/cuda/cuda_fft.cc:467] Unable to register
cuFFT factory: Attempting to register factory for plugin cuFFT when one has
already been registered
WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
E0000 00:00:1772375362.906283      55 cuda_dnn.cc:8579] Unable to register cuDNN
factory: Attempting to register factory for plugin cuDNN when one has already
been registered
E0000 00:00:1772375362.997089      55 cuda_blas.cc:1407] Unable to register
cuBLAS factory: Attempting to register factory for plugin cuBLAS when one has
already been registered
W0000 00:00:1772375363.954536      55 computation_placer.cc:177] computation
placer already registered. Please check linkage and avoid linking the same
target more than once.
W0000 00:00:1772375363.954583      55 computation_placer.cc:177] computation
placer already registered. Please check linkage and avoid linking the same
target more than once.
W0000 00:00:1772375363.954586      55 computation_placer.cc:177] computation
placer already registered. Please check linkage and avoid linking the same
target more than once.
W0000 00:00:1772375363.954589      55 computation_placer.cc:177] computation
placer already registered. Please check linkage and avoid linking the same
target more than once.
```

```
Compute dtype: float16
```

```
TensorFlow: 2.19.0
```

[3]:
```
DATA_DIR = (
    "/kaggle/input/disease-detection-in-rose-leaves"
    "/Disease Detection in Rose Leaves/Dataset/Dataset"
)

# MobileNetV2 native input size (224×224)
IMG_SIZE   = 224
BATCH_SIZE = 16

print(f"Dataset  : {DATA_DIR}")
print(f"IMG_SIZE : {IMG_SIZE}")
print(f"BATCH    : {BATCH_SIZE}")
```

```
Dataset  : /kaggle/input/disease-detection-in-rose-leaves/Disease Detection in
Rose Leaves/Dataset/Dataset
IMG_SIZE : 224
BATCH    : 16
```

1.1 1. Data Loading

Images are loaded from disk, resized to 224×224 (MobileNetV2 native size) and stored as float32 arrays.

[4]:
```
file_paths = []
labels = []

class_names = sorted([
    d for d in os.listdir(DATA_DIR)
    if os.path.isdir(os.path.join(DATA_DIR, d))
])
print("Classes Found:", class_names)

for class_name in class_names:
    class_path = os.path.join(DATA_DIR, class_name)
    for img_name in os.listdir(class_path):
        img_path = os.path.join(class_path, img_name)
        if os.path.isfile(img_path):
            file_paths.append(img_path)
            labels.append(class_name)

file_paths = np.array(file_paths)
labels = np.array(labels)
print(f"Total images discovered: {len(file_paths)}")
```

```
Classes Found: ['Black Spot Leaves', 'Fresh Leaves', 'Hole Leaves', 'Yellow
Leaves']
```

```
Total images discovered: 3366
```

[5]:
```
le = LabelEncoder()
y_encoded = le.fit_transform(labels)
num_classes = len(le.classes_)

print("Encoded Classes:", le.classes_)
print("Number of Classes:", num_classes)
```

```
Encoded Classes: ['Black Spot Leaves' 'Fresh Leaves' 'Hole Leaves' 'Yellow
Leaves']
Number of Classes: 4
```

[6]:
```
x_train_paths, x_temp_paths, y_train, y_temp = train_test_split(
    file_paths, y_encoded,
    test_size=0.2,
    stratify=y_encoded,
    random_state=SEED
)

x_val_paths, x_test_paths, y_val, y_test = train_test_split(
    x_temp_paths, y_temp,
    test_size=0.5,
    stratify=y_temp,
    random_state=SEED
)

print("Train     :", len(x_train_paths))
print("Validation:", len(x_val_paths))
print("Test      :", len(x_test_paths))
```

```
Train     : 2692
Validation: 337
Test      : 337
```

1.2 2. Class Distribution Visualization

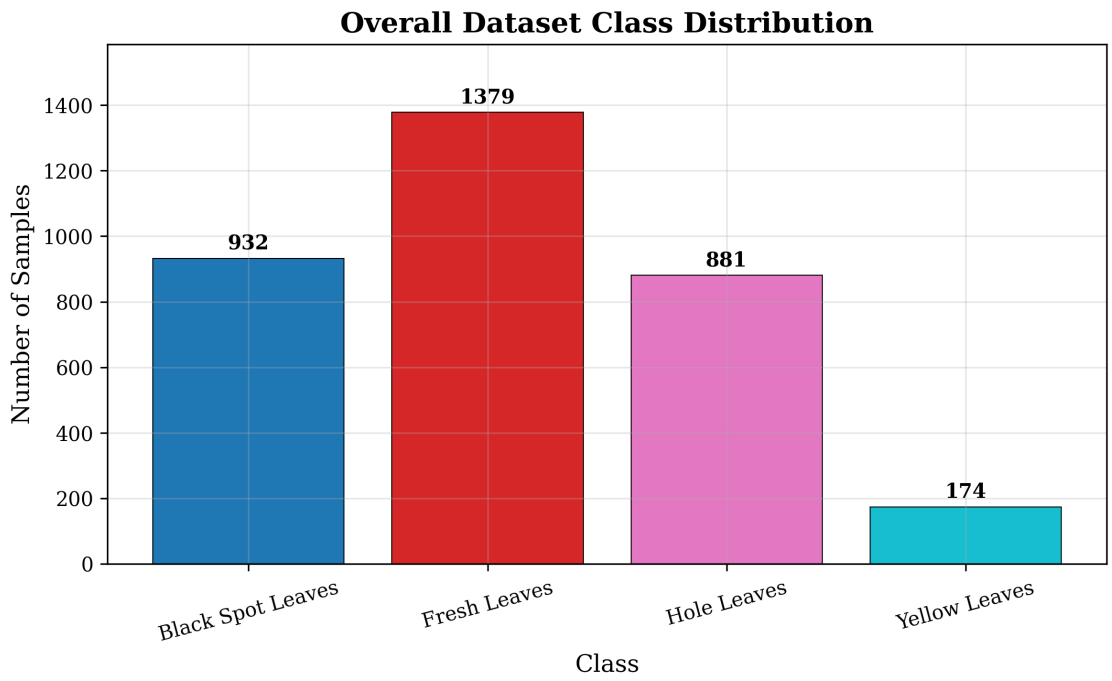
Two publication-quality bar charts: - **Overall distribution** across the full dataset (imbalance check) - **Per-split distribution** (train / validation / test) with grouped bars

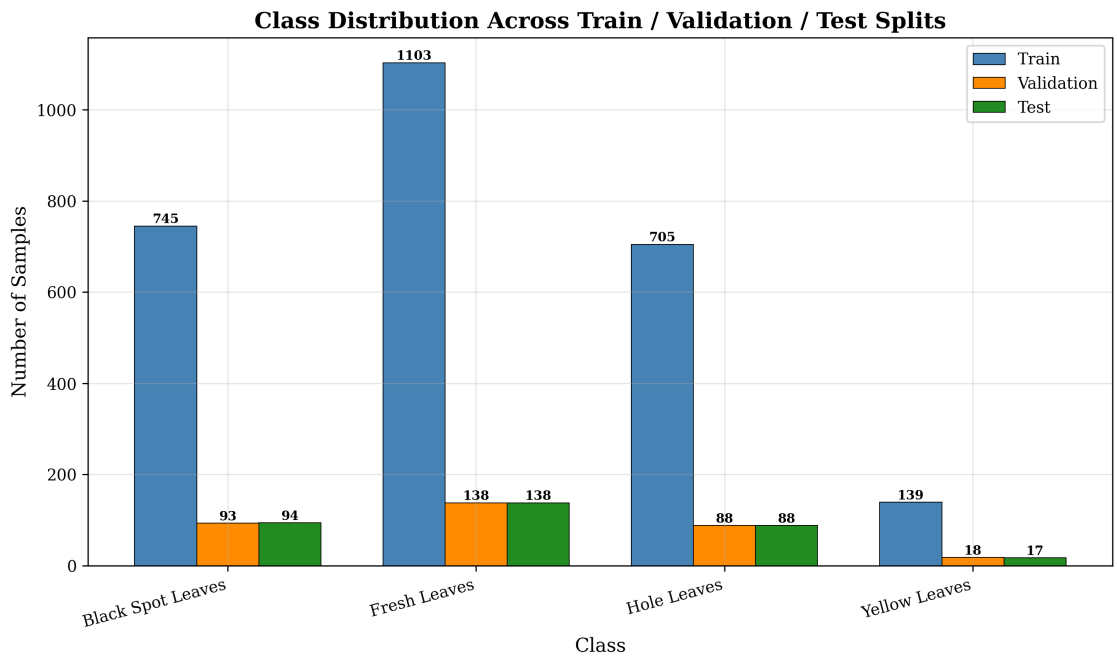
Class imbalance is handled later with **class weights only**.

[7]:
```
#   Color palette (color-blind friendly)
PALETTE = plt.cm.tab10(np.linspace(0, 1, num_classes))

#   2a. Overall class distribution
unique, counts = np.unique(y_encoded, return_counts=True)
class_labels = le.inverse_transform(unique)
```

```python
fig, ax = plt.subplots(figsize=(8, 5))
bars = ax.bar(class_labels, counts, color=PALETTE, edgecolor='black',␣
 ↪linewidth=0.5)
for bar, count in zip(bars, counts):
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + max(counts) * 0.01,
        str(count), ha='center', va='bottom', fontsize=10, fontweight='bold'
    )
ax.set_title('Overall Dataset Class Distribution', fontsize=14,␣
 ↪fontweight='bold')
ax.set_xlabel('Class')
ax.set_ylabel('Number of Samples')
ax.set_ylim(0, max(counts) * 1.15)
ax.tick_params(axis='x', rotation=15)
plt.tight_layout()
plt.savefig('class_distribution.png', dpi=300)
plt.savefig('class_distribution.pdf')
plt.show()

#   2b. Grouped bar chart per split
train_counts = [int(np.sum(y_train == i)) for i in range(num_classes)]
val_counts   = [int(np.sum(y_val   == i)) for i in range(num_classes)]
test_counts  = [int(np.sum(y_test  == i)) for i in range(num_classes)]

x     = np.arange(num_classes)
width = 0.25

fig, ax = plt.subplots(figsize=(10, 6))
b_train = ax.bar(x - width, train_counts, width, label='Train',
                 color='steelblue',   edgecolor='black', linewidth=0.5)
b_val   = ax.bar(x,         val_counts,   width, label='Validation',
                 color='darkorange',  edgecolor='black', linewidth=0.5)
b_test  = ax.bar(x + width, test_counts,  width, label='Test',
                 color='forestgreen', edgecolor='black', linewidth=0.5)

# Annotate each bar with its count
for bars in (b_train, b_val, b_test):
    for bar in bars:
        h = bar.get_height()
        ax.text(
            bar.get_x() + bar.get_width() / 2, h + 0.5,
            str(int(h)), ha='center', va='bottom', fontsize=8, fontweight='bold'
        )

ax.set_xticks(x)
ax.set_xticklabels(le.classes_, rotation=15, ha='right')
```

```
ax.set_xlabel('Class')
ax.set_ylabel('Number of Samples')
ax.set_title('Class Distribution Across Train / Validation / Test Splits',
             fontsize=14, fontweight='bold')
ax.legend()
plt.tight_layout()
plt.savefig('class_distribution_splits.png', dpi=300)
plt.savefig('class_distribution_splits.pdf')
plt.show()
```





1.3 3. Clean tf.data Pipeline + Class Weights

[8]:
```
class_weights = compute_class_weight(
    class_weight='balanced',
    classes=np.unique(y_train),
    y=y_train
)
class_weights = dict(enumerate(class_weights))
print('Class Weights:', class_weights)

AUTOTUNE = tf.data.AUTOTUNE

def decode_image(path, label):
    image = tf.io.read_file(path)
    image = tf.image.decode_jpeg(image, channels=3)
    image = tf.image.resize(image, [IMG_SIZE, IMG_SIZE])
    image = tf.cast(image, tf.float32)
    return image, label


def augment(image, label):
    image = tf.image.random_flip_left_right(image)
    image = tf.image.random_flip_up_down(image)
    image = tf.image.random_brightness(image, max_delta=0.2)
    image = tf.image.random_contrast(image, lower=0.8, upper=1.2)
```

```
    image = tf.image.random_saturation(image, lower=0.75, upper=1.25)
    image = tf.image.random_hue(image, max_delta=0.05)
    image = tf.clip_by_value(image, 0.0, 255.0)
    return image, label


def preprocess_mobilenetv2(image, label):
    image = tf.keras.applications.mobilenet_v2.preprocess_input(image)
    return image, label


train_ds = (
    tf.data.Dataset.from_tensor_slices((x_train_paths, y_train))
    .shuffle(len(x_train_paths), seed=SEED, reshuffle_each_iteration=True)
    .map(decode_image, num_parallel_calls=AUTOTUNE)
    .map(augment, num_parallel_calls=AUTOTUNE)
    .map(preprocess_mobilenetv2, num_parallel_calls=AUTOTUNE)
    .batch(BATCH_SIZE)
    .prefetch(AUTOTUNE)
)

val_ds = (
    tf.data.Dataset.from_tensor_slices((x_val_paths, y_val))
    .map(decode_image, num_parallel_calls=AUTOTUNE)
    .map(preprocess_mobilenetv2, num_parallel_calls=AUTOTUNE)
    .batch(BATCH_SIZE)
    .prefetch(AUTOTUNE)
)

test_ds = (
    tf.data.Dataset.from_tensor_slices((x_test_paths, y_test))
    .map(decode_image, num_parallel_calls=AUTOTUNE)
    .map(preprocess_mobilenetv2, num_parallel_calls=AUTOTUNE)
    .batch(BATCH_SIZE)
    .prefetch(AUTOTUNE)
)

print('TF datasets ready (file-path pipeline, no in-memory full dataset␣
 ↪loading).')
```

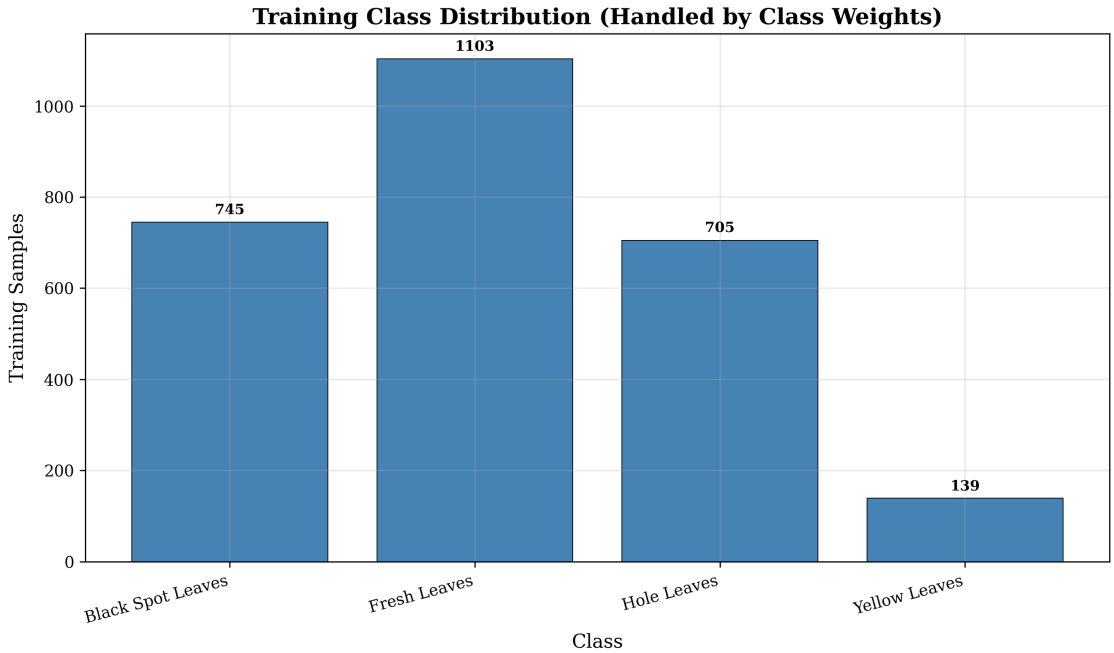
```
Class Weights: {0: np.float64(0.9033557046979865), 1:
np.float64(0.6101541251133273), 2: np.float64(0.9546099290780142), 3:
np.float64(4.841726618705036)}
```

```
I0000 00:00:1772375407.815939      55 gpu_device.cc:2019] Created device
/job:localhost/replica:0/task:0/device:GPU:0 with 15511 MB memory:  -> device:
0, name: Tesla P100-PCIE-16GB, pci bus id: 0000:00:04.0, compute capability: 6.0
```

```
TF datasets ready (file-path pipeline, no in-memory full dataset loading).
```

1.4 3b. Class Weight Summary

[9]:
```
raw_train_counts = np.bincount(y_train, minlength=num_classes)

fig, ax = plt.subplots(figsize=(10, 6))
bars = ax.bar(np.arange(num_classes), raw_train_counts,
              color='steelblue', edgecolor='black', linewidth=0.5)

for i, bar in enumerate(bars):
    h = bar.get_height()
    ax.text(bar.get_x() + bar.get_width() / 2, h + max(raw_train_counts) * 0.01,
            str(int(h)), ha='center', va='bottom', fontsize=9,␣
 ↪fontweight='bold')

ax.set_xticks(np.arange(num_classes))
ax.set_xticklabels(le.classes_, rotation=15, ha='right')
ax.set_xlabel('Class')
ax.set_ylabel('Training Samples')
ax.set_title('Training Class Distribution (Handled by Class Weights)',
             fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('class_weight_distribution.png', dpi=300)
plt.savefig('class_weight_distribution.pdf')
plt.show()

print('\nClass weight summary (single imbalance strategy):')
print(f'  {"Class":<20} {"Count":>10} {"Weight":>10}')
print(f'  {" "*20} {" "*10} {" "*10}')
for i, cls in enumerate(le.classes_):
    print(f'  {cls:<20} {raw_train_counts[i]:>10} {class_weights[i]:>10.4f}')
```




```
Class weight summary (single imbalance strategy):
  Class                     Count     Weight

  Black Spot Leaves           745     0.9034
  Fresh Leaves               1103     0.6102
  Hole Leaves                 705     0.9546
  Yellow Leaves               139     4.8417
```

1.5 4. Model Architecture — MobileNetV2

MobileNetV2 (ImageNet pretrained) is optimised for mobile/edge deployment with a small footprint and efficient depthwise-separable convolutions. Key design decisions: - Backbone layers are partially frozen during Phase 1 (warm-up) then fully unfrozen in Phase 2 (fine-tuning). - **AdamW** optimizer with decoupled weight decay for better regularisation. - **L2** kernel regulariser in the dense head. - **Dropout** in the dense head to prevent overfitting. - Two dense layers for richer feature extraction before the softmax classifier. - Final `Dense` layer uses **float32** output to avoid mixed-precision issues.

[10]:
```
def build_mobilenetv2(config):
    dropout_rate  = float(config['dropout_rate'])
    dense_units   = int(config['dense_units'])
    learning_rate = float(config['learning_rate'])
    weight_decay  = float(config['weight_decay'])
    l2_reg        = float(config['l2_reg'])
    unfreeze_top  = int(config['unfreeze_top'])
```

```
    base = tf.keras.applications.MobileNetV2(
        input_shape=(IMG_SIZE, IMG_SIZE, 3),
        include_top=False,
        weights='imagenet'
    )

    base.trainable = True
    n_to_freeze = max(0, len(base.layers) - unfreeze_top)
    for layer in base.layers[:n_to_freeze]:
        layer.trainable = False

    inputs = tf.keras.Input(shape=(IMG_SIZE, IMG_SIZE, 3))
    x = base(inputs, training=False)
    x = tf.keras.layers.GlobalAveragePooling2D()(x)
    x = tf.keras.layers.BatchNormalization()(x)
    x = tf.keras.layers.Dense(
        dense_units,
        activation='relu',
        kernel_regularizer=tf.keras.regularizers.l2(l2_reg)
    )(x)
    x = tf.keras.layers.Dropout(dropout_rate)(x)
    x = tf.keras.layers.Dense(
        dense_units // 2,
        activation='relu',
        kernel_regularizer=tf.keras.regularizers.l2(l2_reg)
    )(x)
    x = tf.keras.layers.Dropout(dropout_rate / 2)(x)
    x = tf.keras.layers.Activation('linear', dtype='float32')(x)
    outputs = tf.keras.layers.Dense(num_classes, activation='softmax',␣
↪dtype='float32')(x)

    model = tf.keras.Model(inputs, outputs)

    try:
        optimizer = tf.keras.optimizers.AdamW(
            learning_rate=learning_rate, weight_decay=weight_decay
        )
    except (AttributeError, TypeError):
        optimizer = tf.keras.optimizers.Adam(learning_rate=learning_rate)

    model.compile(
        optimizer=optimizer,
        loss='sparse_categorical_crossentropy',
        metrics=['accuracy']
    )
    return model
```

1.6 5. Load Fixed Hyperparameters (Inline, No AutoML)

[11]:
```
best_config = {
    'dropout_rate': 0.3,
    'dense_units': 128,
    'learning_rate': 0.0005161743428272905,
    'weight_decay': 6.426228293037217e-05,
    'l2_reg': 1.9558633566727403e-05,
    'unfreeze_top': 30,
    'optimizer': 'AdamW',
    # Preserved from extracted AutoML output for traceability; compile uses␣
 ↪class-weighted crossentropy.
    'loss': 'sparse_categorical_focal_loss',
    'focal_gamma': 2.0,
    'batch_size': 16,
    'img_size': 224,
}

required_keys = ['dropout_rate', 'dense_units', 'learning_rate',␣
 ↪'weight_decay', 'l2_reg', 'unfreeze_top']
missing = [k for k in required_keys if k not in best_config]
if missing:
    raise ValueError(f'Missing inline hyperparameter keys: {missing}')

if best_config['batch_size'] != BATCH_SIZE or best_config['img_size'] !=␣
 ↪IMG_SIZE:
    raise ValueError(
        f'Inline hyperparameters mismatch:␣
 ↪batch_size={best_config["batch_size"]} (expected {BATCH_SIZE}), '
        f'img_size={best_config["img_size"]} (expected {IMG_SIZE})'
    )

SEP = '=' * 55
print(SEP)
print('  INLINE HYPERPARAMETER CONFIGURATION')
print(SEP)
for key in required_keys:
    print(f'  {key:<13}: {best_config[key]}')
print(SEP)
print('  note: the inline loss field is preserved from extracted HPs; actual␣
 ↪compile path intentionally uses sparse_categorical_crossentropy with class␣
 ↪weights to avoid stacked imbalance bias.')
```

```
=======================================================
  INLINE HYPERPARAMETER CONFIGURATION
=======================================================
  dropout_rate : 0.3
```

```
  dense_units   : 128
  learning_rate: 0.0005161743428272905
  weight_decay : 6.426228293037217e-05
  l2_reg       : 1.9558633566727403e-05
  unfreeze_top : 30
==========================================================
  note: the inline loss field is preserved from extracted HPs; actual compile
path intentionally uses sparse_categorical_crossentropy with class weights to
avoid stacked imbalance bias.
```

1.7 6. Phase 1 — Classifier Head Training

The MobileNetV2 backbone is partially frozen; only the classifier head and the top `unfreeze_top` backbone layers are trained. **ReduceLROnPlateau** halves the LR when validation loss plateaus.

[12]:
```
best_model = build_mobilenetv2(best_config)

callbacks = [
    tf.keras.callbacks.EarlyStopping(
        monitor='val_loss',
        patience=7,
        restore_best_weights=True,
        verbose=1
    ),
    tf.keras.callbacks.ReduceLROnPlateau(
        monitor='val_loss',
        factor=0.3,
        patience=3,
        min_lr=1e-7,
        verbose=1
    ),
    tf.keras.callbacks.ModelCheckpoint(
        'best_rose_model.keras',
        monitor='val_accuracy',
        save_best_only=True,
        verbose=1
    ),
]

print('Phase 1 -- training with partially frozen MobileNetV2...')
history = best_model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=30,
    callbacks=callbacks,
    class_weight=class_weights,
    verbose=1
)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applicatio
ns/mobilenet_v2/mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_224_no_top.h
5
9406464/9406464            1s
0us/step
Phase 1 -- training with partially frozen MobileNetV2…
Epoch 1/30

WARNING: All log messages before absl::InitializeLog() is called are written to
STDERR
I0000 00:00:1772375423.844888     124 service.cc:152] XLA service 0x7c14200110a0
initialized for platform CUDA (this does not guarantee that XLA will be used).
Devices:
I0000 00:00:1772375423.844932     124 service.cc:160]   StreamExecutor device
(0): Tesla P100-PCIE-16GB, Compute Capability 6.0
I0000 00:00:1772375426.176925     124 cuda_dnn.cc:529] Loaded cuDNN version
91002
I0000 00:00:1772375447.774857     124 device_compiler.h:188] Compiled cluster
using XLA!  This line is logged at most once for the lifetime of the process.

169/169            0s 242ms/step -
accuracy: 0.5123 - loss: 1.2456
Epoch 1: val_accuracy improved from -inf to 0.59347, saving model to
best_rose_model.keras
169/169            110s 405ms/step -
accuracy: 0.5127 - loss: 1.2444 - val_accuracy: 0.5935 - val_loss: 1.6139 -
learning_rate: 5.1617e-04
Epoch 2/30
168/169            0s 60ms/step -
accuracy: 0.7111 - loss: 0.7760
Epoch 2: val_accuracy improved from 0.59347 to 0.67656, saving model to
best_rose_model.keras
169/169            12s 70ms/step -
accuracy: 0.7112 - loss: 0.7759 - val_accuracy: 0.6766 - val_loss: 1.9703 -
learning_rate: 5.1617e-04
Epoch 3/30
168/169            0s 59ms/step -
accuracy: 0.7362 - loss: 0.6287
Epoch 3: val_accuracy improved from 0.67656 to 0.71513, saving model to
best_rose_model.keras
169/169            12s 70ms/step -
accuracy: 0.7361 - loss: 0.6291 - val_accuracy: 0.7151 - val_loss: 1.1044 -
learning_rate: 5.1617e-04
Epoch 4/30
168/169            0s 59ms/step -
accuracy: 0.7809 - loss: 0.5702
Epoch 4: val_accuracy improved from 0.71513 to 0.79822, saving model to
best_rose_model.keras
169/169            12s 71ms/step -
```

```
accuracy: 0.7809 - loss: 0.5701 - val_accuracy: 0.7982 - val_loss: 0.8166 -
learning_rate: 5.1617e-04
Epoch 5/30
168/169            0s 59ms/step -
accuracy: 0.8128 - loss: 0.4731
Epoch 5: val_accuracy did not improve from 0.79822
169/169            11s 65ms/step -
accuracy: 0.8128 - loss: 0.4732 - val_accuracy: 0.6706 - val_loss: 2.1846 -
learning_rate: 5.1617e-04
Epoch 6/30
168/169            0s 59ms/step -
accuracy: 0.8191 - loss: 0.4398
Epoch 6: val_accuracy improved from 0.79822 to 0.80119, saving model to
best_rose_model.keras
169/169            12s 70ms/step -
accuracy: 0.8190 - loss: 0.4401 - val_accuracy: 0.8012 - val_loss: 0.7534 -
learning_rate: 5.1617e-04
Epoch 7/30
167/169            0s 59ms/step -
accuracy: 0.8223 - loss: 0.4497
Epoch 7: val_accuracy did not improve from 0.80119
169/169            11s 66ms/step -
accuracy: 0.8224 - loss: 0.4498 - val_accuracy: 0.6083 - val_loss: 1.9689 -
learning_rate: 5.1617e-04
Epoch 8/30
167/169            0s 61ms/step -
accuracy: 0.8466 - loss: 0.3878
Epoch 8: val_accuracy did not improve from 0.80119
169/169            11s 67ms/step -
accuracy: 0.8464 - loss: 0.3880 - val_accuracy: 0.6291 - val_loss: 2.0099 -
learning_rate: 5.1617e-04
Epoch 9/30
169/169            0s 59ms/step -
accuracy: 0.8283 - loss: 0.3840
Epoch 9: ReduceLROnPlateau reducing learning rate to 0.00015485230833292007.

Epoch 9: val_accuracy did not improve from 0.80119
169/169            11s 65ms/step -
accuracy: 0.8283 - loss: 0.3840 - val_accuracy: 0.6320 - val_loss: 1.3389 -
learning_rate: 5.1617e-04
Epoch 10/30
169/169            0s 58ms/step -
accuracy: 0.8630 - loss: 0.3694
Epoch 10: val_accuracy did not improve from 0.80119
169/169            11s 65ms/step -
accuracy: 0.8631 - loss: 0.3691 - val_accuracy: 0.6706 - val_loss: 1.3176 -
learning_rate: 1.5485e-04
Epoch 11/30
```

```
169/169              0s 59ms/step -
accuracy: 0.8830 - loss: 0.2712
Epoch 11: val_accuracy improved from 0.80119 to 0.82196, saving model to
best_rose_model.keras
169/169              12s 70ms/step -
accuracy: 0.8831 - loss: 0.2711 - val_accuracy: 0.8220 - val_loss: 0.7011 -
learning_rate: 1.5485e-04
Epoch 12/30
168/169              0s 59ms/step -
accuracy: 0.9020 - loss: 0.2252
Epoch 12: val_accuracy improved from 0.82196 to 0.82789, saving model to
best_rose_model.keras
169/169              12s 70ms/step -
accuracy: 0.9020 - loss: 0.2254 - val_accuracy: 0.8279 - val_loss: 0.6701 -
learning_rate: 1.5485e-04
Epoch 13/30
168/169              0s 59ms/step -
accuracy: 0.8998 - loss: 0.2255
Epoch 13: val_accuracy improved from 0.82789 to 0.85163, saving model to
best_rose_model.keras
169/169              12s 70ms/step -
accuracy: 0.9001 - loss: 0.2250 - val_accuracy: 0.8516 - val_loss: 0.6373 -
learning_rate: 1.5485e-04
Epoch 14/30
168/169              0s 60ms/step -
accuracy: 0.9331 - loss: 0.1585
Epoch 14: val_accuracy improved from 0.85163 to 0.88131, saving model to
best_rose_model.keras
169/169              12s 70ms/step -
accuracy: 0.9331 - loss: 0.1586 - val_accuracy: 0.8813 - val_loss: 0.3951 -
learning_rate: 1.5485e-04
Epoch 15/30
168/169              0s 59ms/step -
accuracy: 0.9280 - loss: 0.1729
Epoch 15: val_accuracy did not improve from 0.88131
169/169              11s 66ms/step -
accuracy: 0.9280 - loss: 0.1729 - val_accuracy: 0.8546 - val_loss: 0.5382 -
learning_rate: 1.5485e-04
Epoch 16/30
168/169              0s 60ms/step -
accuracy: 0.9460 - loss: 0.1450
Epoch 16: val_accuracy did not improve from 0.88131
169/169              11s 67ms/step -
accuracy: 0.9460 - loss: 0.1452 - val_accuracy: 0.8783 - val_loss: 0.4820 -
learning_rate: 1.5485e-04
Epoch 17/30
168/169              0s 60ms/step -
accuracy: 0.9400 - loss: 0.1596
```

```
Epoch 17: ReduceLROnPlateau reducing learning rate to 4.645569424610585e-05.

Epoch 17: val_accuracy did not improve from 0.88131
169/169            11s 67ms/step -
accuracy: 0.9399 - loss: 0.1601 - val_accuracy: 0.8279 - val_loss: 0.5464 -
learning_rate: 1.5485e-04
Epoch 18/30
169/169            0s 59ms/step -
accuracy: 0.9378 - loss: 0.1686
Epoch 18: val_accuracy improved from 0.88131 to 0.89318, saving model to
best_rose_model.keras
169/169            12s 70ms/step -
accuracy: 0.9378 - loss: 0.1686 - val_accuracy: 0.8932 - val_loss: 0.3681 -
learning_rate: 4.6456e-05
Epoch 19/30
168/169            0s 59ms/step -
accuracy: 0.9504 - loss: 0.1250
Epoch 19: val_accuracy did not improve from 0.89318
169/169            11s 66ms/step -
accuracy: 0.9504 - loss: 0.1251 - val_accuracy: 0.8872 - val_loss: 0.4103 -
learning_rate: 4.6456e-05
Epoch 20/30
168/169            0s 60ms/step -
accuracy: 0.9564 - loss: 0.1088
Epoch 20: val_accuracy improved from 0.89318 to 0.90208, saving model to
best_rose_model.keras
169/169            12s 70ms/step -
accuracy: 0.9563 - loss: 0.1089 - val_accuracy: 0.9021 - val_loss: 0.3563 -
learning_rate: 4.6456e-05
Epoch 21/30
167/169            0s 63ms/step -
accuracy: 0.9569 - loss: 0.1042
Epoch 21: val_accuracy improved from 0.90208 to 0.91098, saving model to
best_rose_model.keras
169/169            21s 74ms/step -
accuracy: 0.9569 - loss: 0.1041 - val_accuracy: 0.9110 - val_loss: 0.3359 -
learning_rate: 4.6456e-05
Epoch 22/30
169/169            0s 59ms/step -
accuracy: 0.9524 - loss: 0.1114
Epoch 22: val_accuracy did not improve from 0.91098
169/169            19s 66ms/step -
accuracy: 0.9524 - loss: 0.1114 - val_accuracy: 0.8991 - val_loss: 0.3437 -
learning_rate: 4.6456e-05
Epoch 23/30
167/169            0s 62ms/step -
accuracy: 0.9505 - loss: 0.1234
Epoch 23: val_accuracy improved from 0.91098 to 0.91691, saving model to
```

```
best_rose_model.keras
169/169              12s 72ms/step -
accuracy: 0.9506 - loss: 0.1232 - val_accuracy: 0.9169 - val_loss: 0.2885 -
learning_rate: 4.6456e-05
Epoch 24/30
168/169              0s 60ms/step -
accuracy: 0.9622 - loss: 0.0967
Epoch 24: val_accuracy did not improve from 0.91691
169/169              11s 66ms/step -
accuracy: 0.9621 - loss: 0.0968 - val_accuracy: 0.9139 - val_loss: 0.2945 -
learning_rate: 4.6456e-05
Epoch 25/30
169/169              0s 59ms/step -
accuracy: 0.9710 - loss: 0.0728
Epoch 25: val_accuracy did not improve from 0.91691
169/169              11s 66ms/step -
accuracy: 0.9710 - loss: 0.0728 - val_accuracy: 0.9050 - val_loss: 0.3526 -
learning_rate: 4.6456e-05
Epoch 26/30
169/169              0s 59ms/step -
accuracy: 0.9678 - loss: 0.0799
Epoch 26: ReduceLROnPlateau reducing learning rate to 1.3936708273831754e-05.

Epoch 26: val_accuracy improved from 0.91691 to 0.91988, saving model to
best_rose_model.keras
169/169              12s 69ms/step -
accuracy: 0.9678 - loss: 0.0799 - val_accuracy: 0.9199 - val_loss: 0.3166 -
learning_rate: 4.6456e-05
Epoch 27/30
169/169              0s 58ms/step -
accuracy: 0.9765 - loss: 0.0770
Epoch 27: val_accuracy improved from 0.91988 to 0.92285, saving model to
best_rose_model.keras
169/169              12s 69ms/step -
accuracy: 0.9765 - loss: 0.0770 - val_accuracy: 0.9228 - val_loss: 0.3068 -
learning_rate: 1.3937e-05
Epoch 28/30
169/169              0s 58ms/step -
accuracy: 0.9641 - loss: 0.0981
Epoch 28: val_accuracy improved from 0.92285 to 0.92582, saving model to
best_rose_model.keras
169/169              20s 69ms/step -
accuracy: 0.9641 - loss: 0.0980 - val_accuracy: 0.9258 - val_loss: 0.3024 -
learning_rate: 1.3937e-05
Epoch 29/30
167/169              0s 59ms/step -
accuracy: 0.9744 - loss: 0.0651
Epoch 29: ReduceLROnPlateau reducing learning rate to 4.181012536719208e-06.
```

```
Epoch 29: val_accuracy improved from 0.92582 to 0.92878, saving model to
best_rose_model.keras
169/169              21s 69ms/step -
accuracy: 0.9744 - loss: 0.0650 - val_accuracy: 0.9288 - val_loss: 0.3098 -
learning_rate: 1.3937e-05
Epoch 30/30
168/169              0s 60ms/step -
accuracy: 0.9752 - loss: 0.0697
Epoch 30: val_accuracy did not improve from 0.92878
169/169              11s 66ms/step -
accuracy: 0.9752 - loss: 0.0697 - val_accuracy: 0.9258 - val_loss: 0.3081 -
learning_rate: 4.1810e-06
Epoch 30: early stopping
Restoring model weights from the end of the best epoch: 23.
```

1.8 7. Phase 2 — Full Fine-tuning

All layers are unfrozen and trained with a cosine decay learning rate schedule to avoid destroying the pretrained ImageNet features while achieving better convergence.

[13]:
```
# Load best Phase-1 checkpoint, then unfreeze the entire backbone
best_model = tf.keras.models.load_model('best_rose_model.keras')

for layer in best_model.layers:
    layer.trainable = True

FINE_TUNE_EPOCHS = 30
steps_per_epoch = (len(x_train_paths) + BATCH_SIZE - 1) // BATCH_SIZE
total_steps = FINE_TUNE_EPOCHS * steps_per_epoch

cosine_schedule = tf.keras.optimizers.schedules.CosineDecay(
    initial_learning_rate=1e-4,
    decay_steps=total_steps,
    alpha=1e-7
)

try:
    ft_optimizer = tf.keras.optimizers.AdamW(
        learning_rate=cosine_schedule, weight_decay=1e-4
    )
except (AttributeError, TypeError):
    ft_optimizer = tf.keras.optimizers.Adam(learning_rate=cosine_schedule)

best_model.compile(
    optimizer=ft_optimizer,
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
```

```
)

ft_callbacks = [
    tf.keras.callbacks.EarlyStopping(
        monitor='val_loss',
        patience=10,
        restore_best_weights=True,
        verbose=1
    ),
    tf.keras.callbacks.ModelCheckpoint(
        'best_rose_model.keras',
        monitor='val_accuracy',
        save_best_only=True,
        verbose=1
    ),
]

print('Phase 2 -- fine-tuning all layers with cosine decay LR schedule...')
history_ft = best_model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=FINE_TUNE_EPOCHS,
    callbacks=ft_callbacks,
    class_weight=class_weights,
    verbose=1
)
```

```
Phase 2 -- fine-tuning all layers with cosine decay LR schedule…
Epoch 1/30
169/169              0s 152ms/step -
accuracy: 0.9668 - loss: 0.0752
Epoch 1: val_accuracy improved from -inf to 0.90801, saving model to
best_rose_model.keras
169/169              73s 266ms/step -
accuracy: 0.9667 - loss: 0.0753 - val_accuracy: 0.9080 - val_loss: 0.3772
Epoch 2/30
169/169              0s 59ms/step -
accuracy: 0.9637 - loss: 0.1170
Epoch 2: val_accuracy did not improve from 0.90801
169/169              11s 67ms/step -
accuracy: 0.9637 - loss: 0.1170 - val_accuracy: 0.9021 - val_loss: 0.4035
Epoch 3/30
169/169              0s 59ms/step -
accuracy: 0.9629 - loss: 0.0934
Epoch 3: val_accuracy improved from 0.90801 to 0.92285, saving model to
best_rose_model.keras
169/169              12s 70ms/step -
```

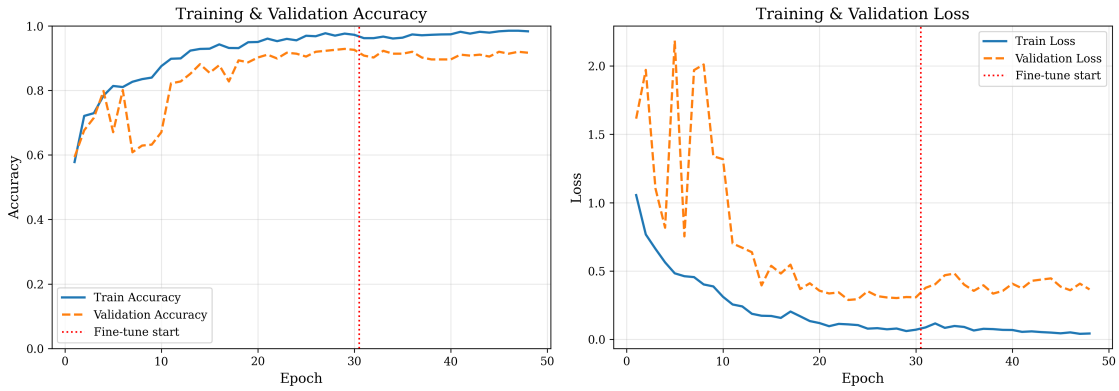
```
accuracy: 0.9630 - loss: 0.0933 - val_accuracy: 0.9228 - val_loss: 0.4694
Epoch 4/30
169/169            0s 59ms/step -
accuracy: 0.9601 - loss: 0.0987
Epoch 4: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9601 - loss: 0.0987 - val_accuracy: 0.9139 - val_loss: 0.4819
Epoch 5/30
169/169            0s 59ms/step -
accuracy: 0.9660 - loss: 0.0897
Epoch 5: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9660 - loss: 0.0897 - val_accuracy: 0.9139 - val_loss: 0.4006
Epoch 6/30
169/169            0s 59ms/step -
accuracy: 0.9736 - loss: 0.0719
Epoch 6: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9736 - loss: 0.0719 - val_accuracy: 0.9199 - val_loss: 0.3552
Epoch 7/30
168/169            0s 60ms/step -
accuracy: 0.9674 - loss: 0.0889
Epoch 7: val_accuracy did not improve from 0.92285
169/169            11s 67ms/step -
accuracy: 0.9675 - loss: 0.0887 - val_accuracy: 0.9021 - val_loss: 0.3958
Epoch 8/30
168/169            0s 60ms/step -
accuracy: 0.9710 - loss: 0.0775
Epoch 8: val_accuracy did not improve from 0.92285
169/169            11s 67ms/step -
accuracy: 0.9710 - loss: 0.0775 - val_accuracy: 0.8961 - val_loss: 0.3349
Epoch 9/30
168/169            0s 59ms/step -
accuracy: 0.9799 - loss: 0.0565
Epoch 9: val_accuracy did not improve from 0.92285
169/169            11s 67ms/step -
accuracy: 0.9798 - loss: 0.0567 - val_accuracy: 0.8961 - val_loss: 0.3547
Epoch 10/30
168/169            0s 59ms/step -
accuracy: 0.9690 - loss: 0.0915
Epoch 10: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9690 - loss: 0.0913 - val_accuracy: 0.8961 - val_loss: 0.4072
Epoch 11/30
168/169            0s 59ms/step -
accuracy: 0.9822 - loss: 0.0477
Epoch 11: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
```

```
accuracy: 0.9822 - loss: 0.0477 - val_accuracy: 0.9110 - val_loss: 0.3733
Epoch 12/30
169/169            0s 59ms/step -
accuracy: 0.9736 - loss: 0.0584
Epoch 12: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9736 - loss: 0.0584 - val_accuracy: 0.9080 - val_loss: 0.4283
Epoch 13/30
168/169            0s 59ms/step -
accuracy: 0.9784 - loss: 0.0605
Epoch 13: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9785 - loss: 0.0605 - val_accuracy: 0.9110 - val_loss: 0.4376
Epoch 14/30
169/169            0s 59ms/step -
accuracy: 0.9791 - loss: 0.0535
Epoch 14: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9791 - loss: 0.0534 - val_accuracy: 0.9050 - val_loss: 0.4463
Epoch 15/30
169/169            0s 59ms/step -
accuracy: 0.9836 - loss: 0.0414
Epoch 15: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9836 - loss: 0.0414 - val_accuracy: 0.9199 - val_loss: 0.3857
Epoch 16/30
168/169            0s 60ms/step -
accuracy: 0.9872 - loss: 0.0550
Epoch 16: val_accuracy did not improve from 0.92285
169/169            11s 67ms/step -
accuracy: 0.9871 - loss: 0.0549 - val_accuracy: 0.9139 - val_loss: 0.3596
Epoch 17/30
168/169            0s 59ms/step -
accuracy: 0.9886 - loss: 0.0324
Epoch 17: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9885 - loss: 0.0325 - val_accuracy: 0.9199 - val_loss: 0.4076
Epoch 18/30
169/169            0s 59ms/step -
accuracy: 0.9793 - loss: 0.0507
Epoch 18: val_accuracy did not improve from 0.92285
169/169            11s 66ms/step -
accuracy: 0.9793 - loss: 0.0506 - val_accuracy: 0.9169 - val_loss: 0.3657
Epoch 18: early stopping
Restoring model weights from the end of the best epoch: 8.
```

1.9 8. Training History

Combined Phase 1 + Phase 2 accuracy and loss curves. The dashed red line marks the transition from Phase 1 to Phase 2.

[14]:
```
phase1_len = len(history.history['accuracy'])

all_acc      = history.history['accuracy']     + history_ft.history['accuracy']
all_val_acc  = history.history['val_accuracy'] + history_ft.
 ↪history['val_accuracy']
all_loss     = history.history['loss']         + history_ft.history['loss']
all_val_loss = history.history['val_loss']     + history_ft.history['val_loss']

epochs_range = range(1, len(all_acc) + 1)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Accuracy
axes[0].plot(epochs_range, all_acc,     label='Train Accuracy',     ␣
 ↪linewidth=2)
axes[0].plot(epochs_range, all_val_acc, label='Validation Accuracy', ␣
 ↪linewidth=2, linestyle='--')
axes[0].axvline(x=phase1_len + 0.5, color='red', linestyle=':', linewidth=1.5,
                label='Fine-tune start')
axes[0].set_title('Training & Validation Accuracy')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('Accuracy')
axes[0].legend()
axes[0].set_ylim(0, 1)

# Loss
axes[1].plot(epochs_range, all_loss,     label='Train Loss',      linewidth=2)
axes[1].plot(epochs_range, all_val_loss, label='Validation Loss',  linewidth=2,␣
 ↪linestyle='--')
axes[1].axvline(x=phase1_len + 0.5, color='red', linestyle=':', linewidth=1.5,
                label='Fine-tune start')
axes[1].set_title('Training & Validation Loss')
axes[1].set_xlabel('Epoch')
axes[1].set_ylabel('Loss')
axes[1].legend()

plt.tight_layout()
plt.savefig('training_history.png', dpi=300)
plt.savefig('training_history.pdf')
plt.show()
```



1.10 9. Model Evaluation

Final evaluation on the held-out test set. Per-class precision, recall, and F1-score are reported via sklearn's `classification_report`.

[15]:
```
best_model = tf.keras.models.load_model('best_rose_model.keras')

loss, acc = best_model.evaluate(test_ds, verbose=1)
print(f"\nTest Accuracy : {acc:.4f}")
print(f"Test Loss     : {loss:.4f}")

y_pred_probs = best_model.predict(test_ds)
pred = np.argmax(y_pred_probs, axis=1)

print("\nClassification Report:")
print(classification_report(y_test, pred, target_names=le.classes_))
```

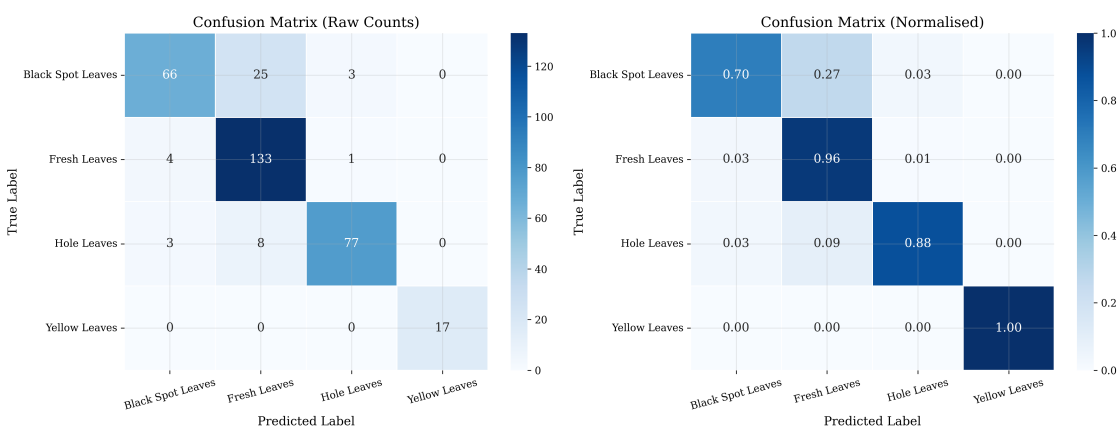
```
22/22              20s 453ms/step -
accuracy: 0.8546 - loss: 0.8287

Test Accuracy : 0.8694
Test Loss     : 0.7136
22/22              19s 455ms/step

Classification Report:
                   precision    recall  f1-score   support

Black Spot Leaves       0.90      0.70      0.79        94
     Fresh Leaves       0.80      0.96      0.88       138
      Hole Leaves       0.95      0.88      0.91        88
    Yellow Leaves       1.00      1.00      1.00        17

         accuracy                           0.87       337
        macro avg       0.91      0.89      0.89       337
```

```
weighted avg       0.88      0.87      0.87       337
```

1.11 10. Confusion Matrix

Left: raw counts. Right: row-normalised (per-class recall on the diagonal).

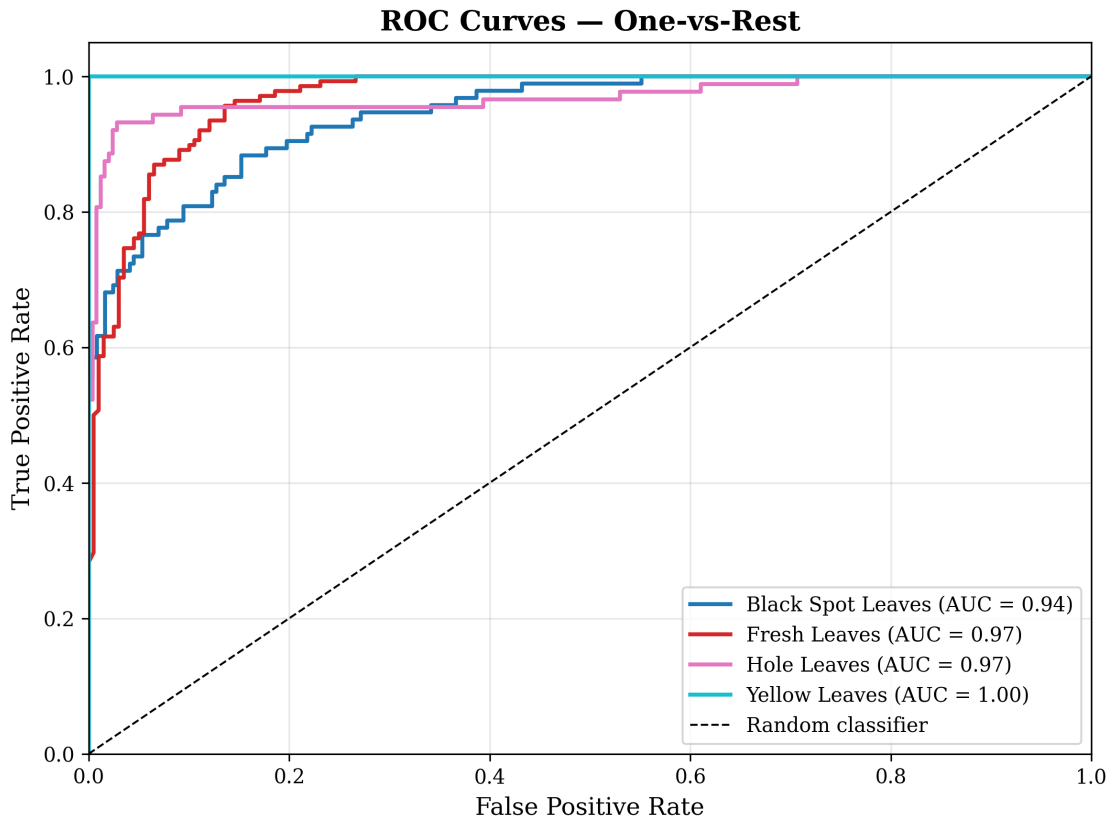
[16]:
```
cm      = confusion_matrix(y_test, pred)
cm_norm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=le.classes_, yticklabels=le.classes_, ax=axes[0],
            linewidths=0.5, linecolor='white')
axes[0].set_xlabel('Predicted Label')
axes[0].set_ylabel('True Label')
axes[0].set_title('Confusion Matrix (Raw Counts)')
axes[0].tick_params(axis='x', rotation=15)
axes[0].tick_params(axis='y', rotation=0)

sns.heatmap(cm_norm, annot=True, fmt='.2f', cmap='Blues',
            xticklabels=le.classes_, yticklabels=le.classes_, ax=axes[1],
            linewidths=0.5, linecolor='white',
            vmin=0, vmax=1)
axes[1].set_xlabel('Predicted Label')
axes[1].set_ylabel('True Label')
axes[1].set_title('Confusion Matrix (Normalised)')
axes[1].tick_params(axis='x', rotation=15)
axes[1].tick_params(axis='y', rotation=0)

plt.tight_layout()
plt.savefig('confusion_matrix.png', dpi=300)
plt.savefig('confusion_matrix.pdf')
plt.show()
```



1.12 11. ROC Curves (One-vs-Rest)

For each class, the true-positive rate vs. false-positive rate curve is computed using a one-vs-rest binarisation. The Area Under the Curve (AUC) is annotated in the legend.

[17]:
```
# Binarise ground truth for one-vs-rest ROC analysis
y_bin = label_binarize(y_test, classes=np.arange(num_classes))

fpr_dict, tpr_dict, roc_auc_dict = {}, {}, {}
for i in range(num_classes):
    fpr_dict[i], tpr_dict[i], _ = roc_curve(y_bin[:, i], y_pred_probs[:, i])
    roc_auc_dict[i] = auc(fpr_dict[i], tpr_dict[i])

colors = plt.cm.tab10(np.linspace(0, 1, num_classes))

fig, ax = plt.subplots(figsize=(8, 6))
for i, color in zip(range(num_classes), colors):
    ax.plot(
        fpr_dict[i], tpr_dict[i], color=color, linewidth=2,
        label=f"{le.classes_[i]} (AUC = {roc_auc_dict[i]:.2f})"
    )
ax.plot([0, 1], [0, 1], 'k--', linewidth=1, label='Random classifier')
ax.set_xlim([0.0, 1.0])
ax.set_ylim([0.0, 1.05])
ax.set_xlabel('False Positive Rate')
ax.set_ylabel('True Positive Rate')
ax.set_title('ROC Curves - One-vs-Rest', fontsize=14, fontweight='bold')
ax.legend(loc='lower right')
plt.tight_layout()
plt.savefig('roc_curves.png', dpi=300)
plt.savefig('roc_curves.pdf')
plt.show()

# Print AUC summary
print('\nAUC per class:')
for i, cls in enumerate(le.classes_):
    print(f'  {cls}: {roc_auc_dict[i]:.4f}')
mean_auc = np.mean(list(roc_auc_dict.values()))
print(f'  Macro-average AUC: {mean_auc:.4f}')
```



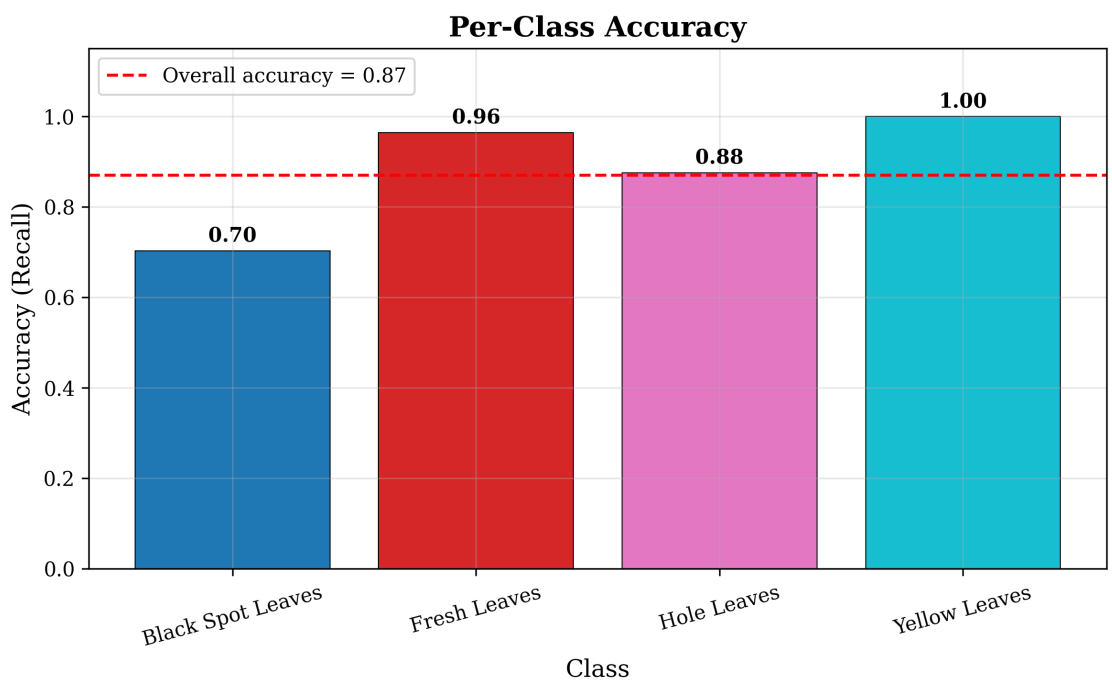

```
AUC per class:
  Black Spot Leaves: 0.9447
  Fresh Leaves: 0.9682
  Hole Leaves: 0.9686
  Yellow Leaves: 1.0000
  Macro-average AUC: 0.9704
```

1.13 12. Per-Class Accuracy

The diagonal of the normalised confusion matrix gives the per-class recall (i.e., accuracy for each class).

[18]:
```
per_class_acc = cm_norm.diagonal()

fig, ax = plt.subplots(figsize=(8, 5))
bar_colors = plt.cm.tab10(np.linspace(0, 1, num_classes))
bars = ax.bar(
    le.classes_, per_class_acc,
    color=bar_colors, edgecolor='black', linewidth=0.5
)
```

```python
for bar, acc_val in zip(bars, per_class_acc):
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.01,
        f'{acc_val:.2f}',
        ha='center', va='bottom', fontsize=10, fontweight='bold'
    )

ax.set_ylim(0, 1.15)
ax.set_ylabel('Accuracy (Recall)')
ax.set_xlabel('Class')
ax.set_title('Per-Class Accuracy', fontsize=14, fontweight='bold')
ax.tick_params(axis='x', rotation=15)
# Add a dashed line at the overall test accuracy
ax.axhline(y=acc, color='red', linestyle='--', linewidth=1.5,
           label=f'Overall accuracy = {acc:.2f}')
ax.legend()
plt.tight_layout()
plt.savefig('per_class_accuracy.png', dpi=300)
plt.savefig('per_class_accuracy.pdf')
plt.show()
```



1.14 13. Bias & Overfitting Validation

This section checks whether yellow-leaf overprediction and overfitting are still visible: - training-vs-validation accuracy gap (generalization gap) - per-class precision/recall from `classification_report` - yellow-class false-positive tendency from confusion matrix columns

[19]:
```
# Combine phase histories for generalization-gap analysis
all_train_acc = np.array(history.history.get('accuracy', []) + history_ft.
 ↪history.get('accuracy', []))
all_val_acc   = np.array(history.history.get('val_accuracy', []) + history_ft.
 ↪history.get('val_accuracy', []))

print('Overfitting check:')
if len(all_train_acc) == 0 or len(all_val_acc) == 0:
    print('    History is empty; cannot compute overfitting gap metrics.')
else:
    best_val_epoch = int(np.argmax(all_val_acc))
    best_val_acc   = float(all_val_acc[best_val_epoch])
    train_at_best  = float(all_train_acc[best_val_epoch])
    gap_at_best    = train_at_best - best_val_acc
    final_gap      = float(all_train_acc[-1] - all_val_acc[-1])

    print(f'  Best val epoch            : {best_val_epoch + 1}')
    print(f'  Train acc at best val     : {train_at_best:.4f}')
    print(f'  Best val acc              : {best_val_acc:.4f}')
    print(f'  Generalization gap (best) : {gap_at_best:.4f}')
    print(f'  Generalization gap (final): {final_gap:.4f}')
    if gap_at_best > 0.10 or final_gap > 0.10:
        print('    Potential overfitting remains (gap > 0.10).')
    else:
        print('    No severe overfitting signal based on accuracy gap threshold.
 ↪')

# Bias check for yellow-leaf overprediction using per-class metrics + confusion␣
 ↪matrix
report = classification_report(y_test, pred, target_names=le.classes_,␣
 ↪output_dict=True)
class_precision = {cls: report[cls]['precision'] for cls in le.classes_}
class_recall = {cls: report[cls]['recall'] for cls in le.classes_}

print('\nPer-class precision/recall:')
for cls in le.classes_:
    print(f'  {cls:<20} precision={class_precision[cls]:.4f} ␣
 ↪recall={class_recall[cls]:.4f}')

yellow_candidates = [cls for cls in le.classes_ if 'yellow' in cls.lower()]
if yellow_candidates:
```

```
    yellow_cls = yellow_candidates[0]
    yi = list(le.classes_).index(yellow_cls)
    # False positives for yellow = all predictions as yellow (column yi) minus␣
↪true yellow predictions.
    yellow_fp = int(cm[:, yi].sum() - cm[yi, yi])
    non_yellow_total = int(cm.sum() - cm[yi, :].sum())

    print(f'\nYellow-bias check for class: {yellow_cls}')
    print(f'  Yellow precision          : {class_precision[yellow_cls]:.4f}')
    print(f'  Yellow recall             : {class_recall[yellow_cls]:.4f}')
    print(f'  Yellow false positives    : {yellow_fp}')

    if non_yellow_total == 0:
        print('  Yellow FP rate (non-yellow): N/A (all evaluation samples are␣
↪yellow)')
    else:
        yellow_fp_rate = yellow_fp / non_yellow_total
        print(f'  Yellow FP rate (non-yellow): {yellow_fp_rate:.4f}')

        if class_precision[yellow_cls] < 0.75 or yellow_fp_rate > 0.15:
            print('    Yellow overprediction risk may still exist; tune␣
↪augmentation/data coverage further.')
        else:
            print('    No strong yellow overprediction signal under current␣
↪thresholds.')
else:
    print('\nNo class containing "yellow" found; skipping yellow-specific bias␣
↪check.')
```

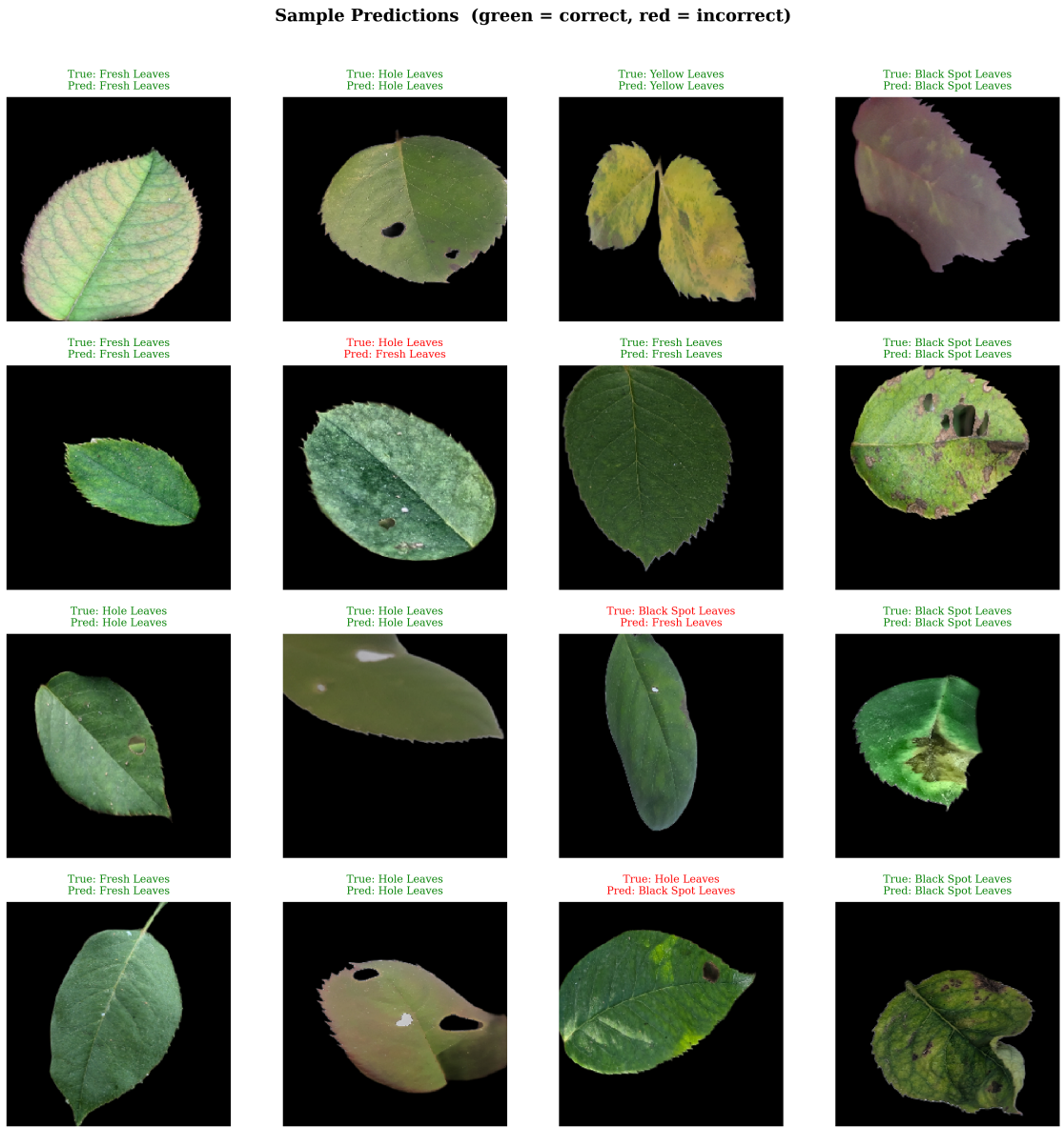
```
Overfitting check:
  Best val epoch            : 29
  Train acc at best val     : 0.9762
  Best val acc              : 0.9288
  Generalization gap (best) : 0.0474
  Generalization gap (final): 0.0664
    No severe overfitting signal based on accuracy gap threshold.

Per-class precision/recall:
  Black Spot Leaves    precision=0.9041  recall=0.7021
  Fresh Leaves         precision=0.8012  recall=0.9638
  Hole Leaves          precision=0.9506  recall=0.8750
  Yellow Leaves        precision=1.0000  recall=1.0000

Yellow-bias check for class: Yellow Leaves
  Yellow precision          : 1.0000
  Yellow recall             : 1.0000
  Yellow false positives    : 0
```

```
Yellow FP rate (non-yellow): 0.0000
  No strong yellow overprediction signal under current thresholds.
```

1.15 14. Sample Predictions Grid

16 random test images are shown with their **true** and **predicted** labels. Green titles indicate correct predictions; red titles indicate errors.

[20]:
```
n_samples = min(16, len(x_test_paths))
rng = np.random.default_rng(SEED)
indices = rng.choice(len(x_test_paths), n_samples, replace=False)

fig, axes = plt.subplots(4, 4, figsize=(12, 12))
axes_flat = axes.flatten()

for ax, i in zip(axes_flat, indices):
    img = tf.keras.utils.load_img(x_test_paths[i], target_size=(IMG_SIZE,␣
 ↪IMG_SIZE))
    img = tf.keras.utils.img_to_array(img).astype(np.uint8)
    true_label = le.classes_[y_test[i]]
    pred_label = le.classes_[pred[i]]
    correct    = (y_test[i] == pred[i])

    ax.imshow(img)
    title_color = 'green' if correct else 'red'
    ax.set_title(
        f'True: {true_label}\nPred: {pred_label}',
        fontsize=8, color=title_color
    )
    ax.axis('off')

for ax in axes_flat[n_samples:]:
    ax.axis('off')

plt.suptitle(
    'Sample Predictions  (green = correct, red = incorrect)',
    fontsize=13, fontweight='bold', y=1.01
)
plt.tight_layout()
plt.savefig('sample_predictions.png', dpi=300, bbox_inches='tight')
plt.savefig('sample_predictions.pdf', bbox_inches='tight')
plt.show()
```



1.16 15. Disease Region Detection & Grad-CAM

Each row shows a sample from one class with three views: 1. **Original** — the input image 2. **Disease regions** — colour-threshold bounding boxes highlighting affected areas 3. **Grad-CAM** — gradient-weighted class activation map showing which regions the model attends to when making its prediction

[21]:
```
def detect_disease_regions(image, class_name):
    """Highlight potential disease regions using colour thresholds in HSV space.
 ↪"""
    img = image.copy()
```

```
    hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
    mask = None

    if 'Black' in class_name:
        lower = np.array([0, 0, 0])
        upper = np.array([180, 255, 80])
        mask = cv2.inRange(hsv, lower, upper)
    elif 'Yellow' in class_name:
        lower = np.array([20, 50, 50])
        upper = np.array([35, 255, 255])
        mask = cv2.inRange(hsv, lower, upper)
    elif 'Hole' in class_name:
        gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
        _, mask = cv2.threshold(gray, 50, 255, cv2.THRESH_BINARY_INV)
    else:
        return img  # Fresh leaves - no disease

    contours, _ = cv2.findContours(mask, cv2.RETR_EXTERNAL, cv2.
↪CHAIN_APPROX_SIMPLE)
    for cnt in contours:
        if cv2.contourArea(cnt) > 200:
            x, y, w, h = cv2.boundingRect(cnt)
            cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 2)
    return img
```

[22]:
```
def make_gradcam_heatmap(img_array, model):
    """
    Compute a Grad-CAM heatmap for the top predicted class.
    Compatible with a Functional model that wraps MobileNetV2 as a sub-model.
    """
    # Locate the MobileNetV2 (or other) sub-model inside the outer model
    base_model = next(
        (l for l in model.layers if isinstance(l, tf.keras.Model)), None
    )
    if base_model is None:
        return None

    # Find the last Conv2D layer inside the backbone
    last_conv = next(
        (l for l in reversed(base_model.layers)
         if isinstance(l, tf.keras.layers.Conv2D)), None
    )
    if last_conv is None:
        return None

    # Build intermediate model exposing conv outputs and backbone output
    base_dual = tf.keras.Model(
```

```
        inputs=base_model.inputs,
        outputs=[last_conv.output, base_model.output]
    )

    # Connect the outer-model head to the backbone via the same layer weights
    outer_input = tf.keras.Input(shape=model.input_shape[1:])
    conv_out_node, mb_out_node = base_dual(outer_input)
    head_layers = [
        l for l in model.layers
        if not isinstance(l, (tf.keras.layers.InputLayer, tf.keras.Model))
    ]
    x = mb_out_node
    for layer in head_layers:
        x = layer(x)
    grad_model = tf.keras.Model(inputs=outer_input, outputs=[conv_out_node, x])

    # Grad-CAM: gradients of the top-class score w.r.t. last conv feature maps
    img_tensor = tf.cast(img_array, tf.float32)
    with tf.GradientTape(watch_accessed_variables=False) as tape:
        tape.watch(img_tensor)
        conv_outputs, predictions = grad_model(img_tensor, training=False)
        tape.watch(conv_outputs)
        pred_index  = tf.argmax(predictions[0])
        class_score = predictions[:, pred_index]

    grads = tape.gradient(class_score, conv_outputs)
    if grads is None:
        return None

    pooled_gradients = tf.reduce_mean(grads, axis=(0, 1, 2))
    heatmap = tf.reduce_sum(conv_outputs[0] * pooled_gradients, axis=-1)
    heatmap = tf.nn.relu(heatmap)
    heatmap = heatmap / (tf.math.reduce_max(heatmap) + 1e-8)
    return heatmap.numpy()


def overlay_gradcam(img_bgr, heatmap, alpha=0.4):
    """Overlay a Grad-CAM heatmap on the original BGR image."""
    h_resized = cv2.resize(heatmap, (img_bgr.shape[1], img_bgr.shape[0]))
    colored   = cv2.applyColorMap(np.uint8(255 * h_resized), cv2.COLORMAP_JET)
    overlay   = cv2.addWeighted(img_bgr, 1 - alpha, colored, alpha, 0)
    return cv2.cvtColor(overlay, cv2.COLOR_BGR2RGB)
```

[23]:
```
fig, axes = plt.subplots(len(class_names), 3,
                         figsize=(12, len(class_names) * 4))
if len(class_names) == 1:
    axes = [axes]  # ensure 2-D indexing for single class
```

```python
for i, class_name in enumerate(class_names):
    class_path = os.path.join(DATA_DIR, class_name)
    img_name   = random.choice(os.listdir(class_path))
    img_path   = os.path.join(class_path, img_name)

    img_bgr = cv2.imread(img_path)
    img_bgr = cv2.resize(img_bgr, (IMG_SIZE, IMG_SIZE))

    # Column 0 - original
    axes[i][0].imshow(cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB))
    axes[i][0].set_title(f'{class_name} (Original)', fontsize=11)
    axes[i][0].axis('off')

    # Column 1 - disease region bounding boxes
    detected = detect_disease_regions(img_bgr, class_name)
    axes[i][1].imshow(cv2.cvtColor(detected, cv2.COLOR_BGR2RGB))
    axes[i][1].set_title(f'{class_name} (Disease Regions)', fontsize=11)
    axes[i][1].axis('off')

    # Column 2 - Grad-CAM
    try:
        img_rgb   = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB).astype('float32')
        processed = tf.keras.applications.mobilenet_v2.preprocess_input(img_rgb.
 ↪copy())
        processed = np.expand_dims(processed, axis=0)
        heatmap   = make_gradcam_heatmap(processed, best_model)
        if heatmap is not None:
            cam_img = overlay_gradcam(img_bgr, heatmap)
            axes[i][2].imshow(cam_img)
            axes[i][2].set_title(f'{class_name} (Grad-CAM)', fontsize=11)
        else:
            axes[i][2].text(0.5, 0.5, 'Grad-CAM N/A',
                            ha='center', va='center', transform=axes[i][2].
 ↪transAxes)
    except Exception as e:
        axes[i][2].text(0.5, 0.5, f'Error: {e}',
                        ha='center', va='center', fontsize=8,
                        transform=axes[i][2].transAxes)
    axes[i][2].axis('off')

plt.tight_layout()
plt.savefig('disease_visualization.png', dpi=300)
plt.savefig('disease_visualization.pdf')
plt.show()
```

Black Spot Leaves (Original)

Black Spot Leaves (Disease Regions)

Error: OpenCV(4.12.0) /io/opencv/modules/imgproc/src/resize.cpp:4086: error: (-215:Assertion failed) func != 0 in function 'resize'

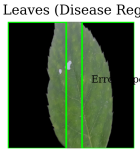
Fresh Leaves (Original)

Fresh Leaves (Disease Regions)

Error: OpenCV(4.12.0) /io/opencv/modules/imgproc/src/resize.cpp:4086: error: (-215:Assertion failed) func != 0 in function 'resize'

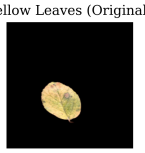
Hole Leaves (Original)

Hole Leaves (Disease Regions)

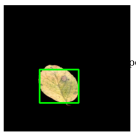


Error: OpenCV(4.12.0) /io/opencv/modules/imgproc/src/resize.cpp:4086: error: (-215:Assertion failed) func != 0 in function 'resize'

Yellow Leaves (Original)



Yellow Leaves (Disease Regions)



Error: OpenCV(4.12.0) /io/opencv/modules/imgproc/src/resize.cpp:4086: error: (-215:Assertion failed) func != 0 in function 'resize'

1.17 16. Save Model

The final model is saved as TFLite (required for Android deployment) in three quantization levels, and also in the Keras v3 format for continued training. A `labels.txt` file is written for Android integration.

[24]:
```
#   Save Keras model for continued training
best_model.save('rose_disease_mobilenetv2.keras')
print('Keras model saved: rose_disease_mobilenetv2.keras')

#   Save class labels for Android integration
with open('labels.txt', 'w') as f:
    for name in le.classes_:
        f.write(name + '\n')
print('Labels saved: labels.txt')

#   a) Float32 TFLite (full precision)
converter = tf.lite.TFLiteConverter.from_keras_model(best_model)
tflite_model = converter.convert()
with open('rose_disease_model.tflite', 'wb') as f:
    f.write(tflite_model)
print(f'Float32 TFLite saved: rose_disease_model.tflite ({len(tflite_model)/1e6:
 ↪.2f} MB)')

#   b) Float16 quantized (~50% smaller)
converter_fp16 = tf.lite.TFLiteConverter.from_keras_model(best_model)
converter_fp16.optimizations = [tf.lite.Optimize.DEFAULT]
converter_fp16.target_spec.supported_types = [tf.float16]
tflite_fp16 = converter_fp16.convert()
with open('rose_disease_model_fp16.tflite', 'wb') as f:
    f.write(tflite_fp16)
print(f'Float16 TFLite saved: rose_disease_model_fp16.tflite ({len(tflite_fp16)/
 ↪1e6:.2f} MB)')

#   c) Int8 fully quantized (~75% smaller, best for low-end devices)
def representative_dataset():
    rep_paths = x_train_paths[:min(200, len(x_train_paths))]
    for i in range(0, len(rep_paths), BATCH_SIZE):
        batch_paths = rep_paths[i:i + BATCH_SIZE]
        batch = []
        for path in batch_paths:
            img = tf.keras.utils.load_img(path, target_size=(IMG_SIZE,␣
 ↪IMG_SIZE))
            img = tf.keras.utils.img_to_array(img)
            img = tf.keras.applications.mobilenet_v2.preprocess_input(img)
            batch.append(img)
        if batch:
            yield [np.stack(batch).astype('float32')]

converter_int8 = tf.lite.TFLiteConverter.from_keras_model(best_model)
converter_int8.optimizations = [tf.lite.Optimize.DEFAULT]
converter_int8.representative_dataset = representative_dataset
converter_int8.target_spec.supported_ops = [tf.lite.OpsSet.TFLITE_BUILTINS_INT8]
```

```
converter_int8.inference_input_type  = tf.uint8
converter_int8.inference_output_type = tf.uint8
tflite_int8 = converter_int8.convert()
with open('rose_disease_model_int8.tflite', 'wb') as f:
    f.write(tflite_int8)
print(f'Int8 TFLite saved: rose_disease_model_int8.tflite ({len(tflite_int8)/
 ↪1e6:.2f} MB)')

best_model.summary(line_length=100)
```

```
Keras model saved: rose_disease_mobilenetv2.keras
Labels saved: labels.txt
INFO:tensorflow:Assets written to: /tmp/tmpu6yb9ove/assets

INFO:tensorflow:Assets written to: /tmp/tmpu6yb9ove/assets

Saved artifact at '/tmp/tmpu6yb9ove'. The following endpoints are available:

* Endpoint 'serve'
  args_0 (POSITIONAL_ONLY): TensorSpec(shape=(None, 224, 224, 3),
dtype=tf.float32, name='input_layer_1')
Output Type:
  TensorSpec(shape=(None, 4), dtype=tf.float32, name=None)
Captures:
  136426350611728: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350613072: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350612880: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350612496: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350613648: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350611920: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350613264: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350613456: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350610768: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350614416: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350613840: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350614032: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350614224: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350612304: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350615376: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350614800: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350614992: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350615184: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350612688: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350616336: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350615760: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350615952: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350616144: TensorSpec(shape=(), dtype=tf.resource, name=None)
  136426350612112: TensorSpec(shape=(), dtype=tf.resource, name=None)
```

```
136426350617296: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350616720: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350616912: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350617104: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350614608: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350618256: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350617680: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350617872: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350618064: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350615568: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350619216: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350618640: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350618832: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350619024: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350616528: TensorSpec(shape=(), dtype=tf.resource, name=None)
136426350620176: TensorSpec(shape=(), dtype=tf.resource, name=None)
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W0000 00:00:1772376253.488189      55 tf_tfl_flatbuffer_helpers.cc:365] Ignored
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W0000 00:00:1772376253.488224      55 tf_tfl_flatbuffer_helpers.cc:368] Ignored
drop_control_dependency.
I0000 00:00:1772376253.639998      55 mlir_graph_optimization_pass.cc:425] MLIR
V1 optimization pass is not enabled
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'tf.Relu6' op is neither a custom op nor a flex op
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fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
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'tf.Conv2D' op is neither a custom op nor a flex op
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fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
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'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["AddV2:", "functional_1/mobilenetv2_1.00_224_1/block
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.AddV2' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
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'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
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fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Pad' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["DepthwiseConv2dNative:", "functional_1/mobilenetv2_
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.DepthwiseConv2dNative' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
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loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
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fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
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k_12_project_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["AddV2:", "functional_1/mobilenetv2_1.00_224_1/block
_12_add_1/Add@__inference_function_208818"] at fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.AddV2' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_13_expand_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_13_expand_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Pad:", "functional_1/mobilenetv2_1.00_224_1/block_1
3_pad_1/Pad@__inference_function_208818"] at fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Pad' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["DepthwiseConv2dNative:", "functional_1/mobilenetv2_
1.00_224_1/block_13_depthwise_1/depthwise@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.DepthwiseConv2dNative' op is neither a custom op nor a flex op
```

```
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_13_depthwise_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_13_project_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_14_expand_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_14_expand_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["DepthwiseConv2dNative:", "functional_1/mobilenetv2_
1.00_224_1/block_14_depthwise_1/depthwise@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.DepthwiseConv2dNative' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_14_depthwise_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_14_project_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["AddV2:", "functional_1/mobilenetv2_1.00_224_1/block
_14_add_1/Add@__inference_function_208818"] at fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.AddV2' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
```

```
k_15_expand_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_15_expand_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["DepthwiseConv2dNative:", "functional_1/mobilenetv2_
1.00_224_1/block_15_depthwise_1/depthwise@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.DepthwiseConv2dNative' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_15_depthwise_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_15_project_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["AddV2:", "functional_1/mobilenetv2_1.00_224_1/block
_15_add_1/Add@__inference_function_208818"] at fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.AddV2' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_16_expand_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_16_expand_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["DepthwiseConv2dNative:", "functional_1/mobilenetv2_
1.00_224_1/block_16_depthwise_1/depthwise@__inference_function_208818"] at
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```
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.DepthwiseConv2dNative' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/block
_16_depthwise_relu_1/Relu6@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/bloc
k_16_project_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/Conv
_1_1/convolution@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Conv2D' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu6:", "functional_1/mobilenetv2_1.00_224_1/out_r
elu_1/Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu6' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["MatMul:",
"functional_1/dense_1/MatMul@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.MatMul' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["BiasAdd:",
"functional_1/dense_1/BiasAdd@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.BiasAdd' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu:",
"functional_1/dense_1/Relu@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["MatMul:",
"functional_1/dense_1_2/MatMul@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
```

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"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.MatMul' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["BiasAdd:",
"functional_1/dense_1_2/BiasAdd@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.BiasAdd' op is neither a custom op nor a flex op
loc(callsite(callsite(fused["Relu:",
"functional_1/dense_1_2/Relu@__inference_function_208818"] at
fused["StatefulPartitionedCall:",
"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at
fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): error:
'tf.Relu' op is neither a custom op nor a flex op
error: failed while converting: 'main':
Some ops are not supported by the native TFLite runtime, you can enable TF
kernels fallback using TF Select. See instructions:
https://www.tensorflow.org/lite/guide/ops_select
TF Select ops: AddV2, BiasAdd, Conv2D, DepthwiseConv2dNative, MatMul, Pad, Relu,
Relu6
Details:
        tf.AddV2(tensor<?x14x14x64xf16>, tensor<?x14x14x64xf16>) ->
(tensor<?x14x14x64xf16>) : {device = ""}
        tf.AddV2(tensor<?x14x14x96xf16>, tensor<?x14x14x96xf16>) ->
(tensor<?x14x14x96xf16>) : {device = ""}
        tf.AddV2(tensor<?x28x28x32xf16>, tensor<?x28x28x32xf16>) ->
(tensor<?x28x28x32xf16>) : {device = ""}
        tf.AddV2(tensor<?x56x56x24xf16>, tensor<?x56x56x24xf16>) ->
(tensor<?x56x56x24xf16>) : {device = ""}
        tf.AddV2(tensor<?x7x7x160xf16>, tensor<?x7x7x160xf16>) ->
(tensor<?x7x7x160xf16>) : {device = ""}
        tf.BiasAdd(tensor<?x128xf16>, tensor<128xf16>) -> (tensor<?x128xf16>) :
{data_format = "NHWC", device = ""}
        tf.BiasAdd(tensor<?x64xf16>, tensor<64xf16>) -> (tensor<?x64xf16>) :
{data_format = "NHWC", device = ""}
        tf.Conv2D(tensor<?x112x112x16xf16>, tensor<1x1x16x96xf16>) ->
(tensor<?x112x112x96xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x112x112x32xf16>, tensor<1x1x32x16xf16>) ->
(tensor<?x112x112x16xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x192xf16>, tensor<1x1x192x64xf16>) ->
(tensor<?x14x14x64xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
```

```
        tf.Conv2D(tensor<?x14x14x384xf16>, tensor<1x1x384x64xf16>) ->
(tensor<?x14x14x64xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x384xf16>, tensor<1x1x384x96xf16>) ->
(tensor<?x14x14x96xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x576xf16>, tensor<1x1x576x96xf16>) ->
(tensor<?x14x14x96xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x64xf16>, tensor<1x1x64x384xf16>) ->
(tensor<?x14x14x384xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x96xf16>, tensor<1x1x96x576xf16>) ->
(tensor<?x14x14x576xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x224x224x3xf16>, tensor<3x3x3x32xf16>) ->
(tensor<?x112x112x32xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 2, 2, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x28x28x144xf16>, tensor<1x1x144x32xf16>) ->
(tensor<?x28x28x32xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x28x28x192xf16>, tensor<1x1x192x32xf16>) ->
(tensor<?x28x28x32xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x28x28x32xf16>, tensor<1x1x32x192xf16>) ->
(tensor<?x28x28x192xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x56x56x144xf16>, tensor<1x1x144x24xf16>) ->
(tensor<?x56x56x24xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x56x56x24xf16>, tensor<1x1x24x144xf16>) ->
(tensor<?x56x56x144xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x56x56x96xf16>, tensor<1x1x96x24xf16>) ->
(tensor<?x56x56x24xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
```

```
        tf.Conv2D(tensor<?x7x7x160xf16>, tensor<1x1x160x960xf16>) ->
(tensor<?x7x7x960xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1,
1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x7x7x320xf16>, tensor<1x1x320x1280xf16>) ->
(tensor<?x7x7x1280xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x7x7x576xf16>, tensor<1x1x576x160xf16>) ->
(tensor<?x7x7x160xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1,
1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x7x7x960xf16>, tensor<1x1x960x160xf16>) ->
(tensor<?x7x7x160xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1,
1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x7x7x960xf16>, tensor<1x1x960x320xf16>) ->
(tensor<?x7x7x320xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1,
1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],
use_cudnn_on_gpu = true}
        tf.DepthwiseConv2dNative(tensor<?x112x112x32xf16>, tensor<3x3x32x1xf16>)
-> (tensor<?x112x112x32xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}
        tf.DepthwiseConv2dNative(tensor<?x113x113x96xf16>, tensor<3x3x96x1xf16>)
-> (tensor<?x56x56x96xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2, 1]}
        tf.DepthwiseConv2dNative(tensor<?x14x14x384xf16>, tensor<3x3x384x1xf16>)
-> (tensor<?x14x14x384xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}
        tf.DepthwiseConv2dNative(tensor<?x14x14x576xf16>, tensor<3x3x576x1xf16>)
-> (tensor<?x14x14x576xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}
        tf.DepthwiseConv2dNative(tensor<?x15x15x576xf16>, tensor<3x3x576x1xf16>)
-> (tensor<?x7x7x576xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2, 1]}
        tf.DepthwiseConv2dNative(tensor<?x28x28x192xf16>, tensor<3x3x192x1xf16>)
-> (tensor<?x28x28x192xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}
        tf.DepthwiseConv2dNative(tensor<?x29x29x192xf16>, tensor<3x3x192x1xf16>)
-> (tensor<?x14x14x192xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2, 1]}
        tf.DepthwiseConv2dNative(tensor<?x56x56x144xf16>, tensor<3x3x144x1xf16>)
-> (tensor<?x56x56x144xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}
        tf.DepthwiseConv2dNative(tensor<?x57x57x144xf16>, tensor<3x3x144x1xf16>)
-> (tensor<?x28x28x144xf16>) : {data_format = "NHWC", device = "", dilations =
[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2, 1]}
        tf.DepthwiseConv2dNative(tensor<?x7x7x960xf16>, tensor<3x3x960x1xf16>)
```

```
-> (tensor<?x7x7x960xf16>) : {data_format = "NHWC", device = "", dilations = [1,
1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}
        tf.MatMul(tensor<?x1280xf16>, tensor<128x1280xf16>) ->
(tensor<?x128xf16>) : {grad_a = false, grad_b = false, transpose_a = false,
transpose_b = true}
        tf.MatMul(tensor<?x128xf16>, tensor<64x128xf16>) -> (tensor<?x64xf16>) :
{grad_a = false, grad_b = false, transpose_a = false, transpose_b = true}
        tf.Pad(tensor<?x112x112x96xf16>, tensor<4x2xi32>) ->
(tensor<?x113x113x96xf16>) : {device = ""}
        tf.Pad(tensor<?x14x14x576xf16>, tensor<4x2xi32>) ->
(tensor<?x15x15x576xf16>) : {device = ""}
        tf.Pad(tensor<?x28x28x192xf16>, tensor<4x2xi32>) ->
(tensor<?x29x29x192xf16>) : {device = ""}
        tf.Pad(tensor<?x56x56x144xf16>, tensor<4x2xi32>) ->
(tensor<?x57x57x144xf16>) : {device = ""}
        tf.Relu(tensor<?x128xf16>) -> (tensor<?x128xf16>) : {device = ""}
        tf.Relu(tensor<?x64xf16>) -> (tensor<?x64xf16>) : {device = ""}
        tf.Relu6(tensor<?x112x112x32xf16>) -> (tensor<?x112x112x32xf16>) :
{device = ""}
        tf.Relu6(tensor<?x112x112x96xf16>) -> (tensor<?x112x112x96xf16>) :
{device = ""}
        tf.Relu6(tensor<?x14x14x192xf16>) -> (tensor<?x14x14x192xf16>) : {device
= ""}
        tf.Relu6(tensor<?x14x14x384xf16>) -> (tensor<?x14x14x384xf16>) : {device
= ""}
        tf.Relu6(tensor<?x14x14x576xf16>) -> (tensor<?x14x14x576xf16>) : {device
= ""}
        tf.Relu6(tensor<?x28x28x144xf16>) -> (tensor<?x28x28x144xf16>) : {device
= ""}
        tf.Relu6(tensor<?x28x28x192xf16>) -> (tensor<?x28x28x192xf16>) : {device
= ""}
        tf.Relu6(tensor<?x56x56x144xf16>) -> (tensor<?x56x56x144xf16>) : {device
= ""}
        tf.Relu6(tensor<?x56x56x96xf16>) -> (tensor<?x56x56x96xf16>) : {device =
""}
        tf.Relu6(tensor<?x7x7x1280xf16>) -> (tensor<?x7x7x1280xf16>) : {device =
""}
        tf.Relu6(tensor<?x7x7x576xf16>) -> (tensor<?x7x7x576xf16>) : {device =
""}
        tf.Relu6(tensor<?x7x7x960xf16>) -> (tensor<?x7x7x960xf16>) : {device =
""}
```

```
---------------------------------------------------------------------------
ConverterError                            Traceback (most recent call last)
/tmp/ipykernel_55/652155499.py in <cell line: 0>()
     11 #   a) Float32 TFLite (full precision)
```

```
     12 converter = tf.lite.TFLiteConverter.from_keras_model(best_model)
---> 13 tflite_model = converter.convert()
     14 with open('rose_disease_model.tflite', 'wb') as f:
     15     f.write(tflite_model)

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/lite.py in␣
↪wrapper(self, *args, **kwargs)
   1248   def wrapper(self, *args, **kwargs):
   1249     # pylint: disable=protected-access
-> 1250     return self._convert_and_export_metrics(convert_func, *args,␣
↪**kwargs)
   1251     # pylint: enable=protected-access
   1252

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/lite.py in␣
↪_convert_and_export_metrics(self, convert_func, *args, **kwargs)
   1200     self._save_conversion_params_metric()
   1201     start_time = time.process_time()
-> 1202     result = convert_func(self, *args, **kwargs)
   1203     elapsed_time_ms = (time.process_time() - start_time) * 1000
   1204     if result:

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/lite.py in␣
↪convert(self)
   1766         Invalid quantization parameters.
   1767     """
-> 1768     saved_model_convert_result = self._convert_as_saved_model()
   1769     if saved_model_convert_result:
   1770       return saved_model_convert_result

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/lite.py in␣
↪_convert_as_saved_model(self)
   1747       )
   1748       if self.saved_model_dir:
-> 1749         return super(TFLiteKerasModelConverterV2, self).convert(

   1750             graph_def, input_tensors, output_tensors
   1751         )

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/lite.py in␣
↪convert(self, graph_def, input_tensors, output_tensors)
   1485
   1486     # Converts model.
-> 1487     result = _convert_graphdef(

   1488         input_data=graph_def,
   1489         input_tensors=input_tensors,
```

```
/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/convert_phase.py␣
↪in wrapper(*args, **kwargs)
    210         else:
    211           report_error_message(str(converter_error))
--> 212         raise converter_error from None  # Re-throws the exception.
    213       except Exception as error:
    214         report_error_message(str(error))

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/convert_phase.py␣
↪in wrapper(*args, **kwargs)
    203     def wrapper(*args, **kwargs):
    204       try:
--> 205         return func(*args, **kwargs)
    206       except ConverterError as converter_error:
    207         if converter_error.errors:

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/convert.py in␣
↪convert_graphdef(input_data, input_tensors, output_tensors, **kwargs)
    883       model_flags.output_arrays.append(util.
↪get_tensor_name(output_tensor))
    884
--> 885   data = convert(
    886       model_flags,
    887       conversion_flags,

/usr/local/lib/python3.12/dist-packages/tensorflow/lite/python/convert.py in␣
↪convert(model_flags, conversion_flags, input_data_str, debug_info_str)
    348             debug_info_str,
    349         )
--> 350     raise converter_error
    351
    352


ConverterError: Could not translate MLIR to FlatBuffer.<unknown>:0: error:␣
↪loc(callsite(callsite(fused["Conv2D:", "functional_1/mobilenetv2_1.00_224_1/
↪Conv1_1/convolution@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv1_1/convolution@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %167 = "tf.Conv2D"(%166, %156) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 2, 2,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x224x224x3xf16>,␣
↪tensor<3x3x3x32xf16>) -> tensor<?x112x112x32xf16>
```

```
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv1_1/convolution@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv1_relu_1/Relu6@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv1_relu_1/Relu6@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %172 = "tf.Relu6"(%171) {device = ""} : (tensor<?x112x112x32xf16>)␣
↪-> tensor<?x112x112x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv1_relu_1/Relu6@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/expanded_conv_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/expanded_conv_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %173 = "tf.DepthwiseConv2dNative"(%172, %160) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x112x112x32xf16>,␣
↪tensor<3x3x32x1xf16>) -> tensor<?x112x112x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/expanded_conv_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/expanded_conv_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

```
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/expanded_conv_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %178 = "tf.Relu6"(%177) {device = ""} : (tensor<?x112x112x32xf16>)
↪-> tensor<?x112x112x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/expanded_conv_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/expanded_conv_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/expanded_conv_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %179 = "tf.Conv2D"(%178, %163) <{data_format = "NHWC", dilations =
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x112x112x32xf16>,␣
↪tensor<1x1x32x16xf16>) -> tensor<?x112x112x16xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/expanded_conv_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %184 = "tf.Conv2D"(%183, %76) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x112x112x16xf16>,␣
↪tensor<1x1x16x96xf16>) -> tensor<?x112x112x96xf16>
```

```
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %189 = "tf.Relu6"(%188) {device = ""} : (tensor<?x112x112x96xf16>) ␣
↪-> tensor<?x112x112x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Pad' op ␣
↪is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %190 = "tf.Pad"(%189, %5) {device = ""} : (tensor<?
↪x112x112x96xf16>, tensor<4x2xi32>) -> tensor<?x113x113x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_1_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

```
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_1_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %191 = "tf.DepthwiseConv2dNative"(%190, %73) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "VALID",␣
↪strides = [1, 2, 2, 1]}> {device = ""} : (tensor<?x113x113x96xf16>,␣
↪tensor<3x3x96x1xf16>) -> tensor<?x56x56x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_1_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %196 = "tf.Relu6"(%195) {device = ""} : (tensor<?x56x56x96xf16>) -␣
↪tensor<?x56x56x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %197 = "tf.Conv2D"(%196, %79) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x96xf16>,␣
↪tensor<1x1x96x24xf16>) -> tensor<?x56x56x24xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_1_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %202 = "tf.Conv2D"(%201, %85) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x24xf16>,␣
↪tensor<1x1x24x144xf16>) -> tensor<?x56x56x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %207 = "tf.Relu6"(%206) {device = ""} : (tensor<?x56x56x144xf16>)␣
↪-> tensor<?x56x56x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_2_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_2_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %208 = "tf.DepthwiseConv2dNative"(%207, %82) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x56x56x144xf16>,␣
↪tensor<3x3x144x1xf16>) -> tensor<?x56x56x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_2_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %213 = "tf.Relu6"(%212) {device = ""} : (tensor<?x56x56x144xf16>)␣
↪-> tensor<?x56x56x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %214 = "tf.Conv2D"(%213, %88) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x144xf16>,␣
↪tensor<1x1x144x24xf16>) -> tensor<?x56x56x24xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %219 = "tf.AddV2"(%201, %218) {device = ""} : (tensor<?
↪x56x56x24xf16>, tensor<?x56x56x24xf16>) -> tensor<?x56x56x24xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_2_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %220 = "tf.Conv2D"(%219, %94) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x24xf16>,␣
↪tensor<1x1x24x144xf16>) -> tensor<?x56x56x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %225 = "tf.Relu6"(%224) {device = ""} : (tensor<?x56x56x144xf16>)␣
↪-> tensor<?x56x56x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Pad' op
↪is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %226 = "tf.Pad"(%225, %5) {device = ""} : (tensor<?x56x56x144xf16>
↪tensor<4x2xi32>) -> tensor<?x57x57x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_3_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_3_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %227 = "tf.DepthwiseConv2dNative"(%226, %91) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "VALID",␣
↪strides = [1, 2, 2, 1]}> {device = ""} : (tensor<?x57x57x144xf16>,␣
↪tensor<3x3x144x1xf16>) -> tensor<?x28x28x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_3_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
```

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<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %232 = "tf.Relu6"(%231) {device = ""} : (tensor<?x28x28x144xf16>)␣
↪-> tensor<?x28x28x144xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:"
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:"
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %233 = "tf.Conv2D"(%232, %97) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x144xf16>,␣
↪tensor<1x1x144x32xf16>) -> tensor<?x28x28x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_3_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:"
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:"
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %238 = "tf.Conv2D"(%237, %103) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x32xf16>,␣
↪tensor<1x1x32x192xf16>) -> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %243 = "tf.Relu6"(%242) {device = ""} : (tensor<?x28x28x192xf16>)␣
↪-> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_4_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_4_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %244 = "tf.DepthwiseConv2dNative"(%243, %100) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x28x28x192xf16>,␣
↪tensor<3x3x192x1xf16>) -> tensor<?x28x28x192xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_4_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %249 = "tf.Relu6"(%248) {device = ""} : (tensor<?x28x28x192xf16>)␣
↪-> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %250 = "tf.Conv2D"(%249, %106) <{data_format = "NHWC", dilations =
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x192xf16>,␣
↪tensor<1x1x192x32xf16>) -> tensor<?x28x28x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %255 = "tf.AddV2"(%237, %254) {device = ""} : (tensor<?
↪x28x28x32xf16>, tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_4_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %256 = "tf.Conv2D"(%255, %112) <{data_format = "NHWC", dilations =
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x32xf16>,␣
↪tensor<1x1x32x192xf16>) -> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %261 = "tf.Relu6"(%260) {device = ""} : (tensor<?x28x28x192xf16>)␣
↪-> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_5_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_5_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %262 = "tf.DepthwiseConv2dNative"(%261, %109) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x28x28x192xf16>,␣
↪tensor<3x3x192x1xf16>) -> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_5_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %267 = "tf.Relu6"(%266) {device = ""} : (tensor<?x28x28x192xf16>)␣
↪-> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %268 = "tf.Conv2D"(%267, %115) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x192xf16>,␣
↪tensor<1x1x192x32xf16>) -> tensor<?x28x28x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %273 = "tf.AddV2"(%255, %272) {device = ""} : (tensor<?
↪x28x28x32xf16>, tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_5_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %274 = "tf.Conv2D"(%273, %121) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x32xf16>,␣
↪tensor<1x1x32x192xf16>) -> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
```

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<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %279 = "tf.Relu6"(%278) {device = ""} : (tensor<?x28x28x192xf16>)␣
↪-> tensor<?x28x28x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Pad' op␣
↪is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %280 = "tf.Pad"(%279, %5) {device = ""} : (tensor<?x28x28x192xf16>␣
↪tensor<4x2xi32>) -> tensor<?x29x29x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_6_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
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<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_6_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %281 = "tf.DepthwiseConv2dNative"(%280, %118) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "VALID",␣
↪strides = [1, 2, 2, 1]}> {device = ""} : (tensor<?x29x29x192xf16>,␣
↪tensor<3x3x192x1xf16>) -> tensor<?x14x14x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_6_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %286 = "tf.Relu6"(%285) {device = ""} : (tensor<?x14x14x192xf16>)␣
↪-> tensor<?x14x14x192xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %287 = "tf.Conv2D"(%286, %124) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x192xf16>,␣
↪tensor<1x1x192x64xf16>) -> tensor<?x14x14x64xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_6_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %292 = "tf.Conv2D"(%291, %130) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %297 = "tf.Relu6"(%296) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_7_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
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<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_7_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %298 = "tf.DepthwiseConv2dNative"(%297, %127) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_7_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %303 = "tf.Relu6"(%302) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %304 = "tf.Conv2D"(%303, %133) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x64xf16>) -> tensor<?x14x14x64xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %309 = "tf.AddV2"(%291, %308) {device = ""} : (tensor<?
↪x14x14x64xf16>, tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_7_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %310 = "tf.Conv2D"(%309, %139) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
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<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %315 = "tf.Relu6"(%314) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_8_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_8_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %316 = "tf.DepthwiseConv2dNative"(%315, %136) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_8_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %321 = "tf.Relu6"(%320) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
```

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<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %322 = "tf.Conv2D"(%321, %142) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x64xf16>) -> tensor<?x14x14x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %327 = "tf.AddV2"(%309, %326) {device = ""} : (tensor<?
↪x14x14x64xf16>, tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_8_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %328 = "tf.Conv2D"(%327, %148) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %333 = "tf.Relu6"(%332) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_9_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_9_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %334 = "tf.DepthwiseConv2dNative"(%333, %145) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_9_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %339 = "tf.Relu6"(%338) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %340 = "tf.Conv2D"(%339, %151) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x64xf16>) -> tensor<?x14x14x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %345 = "tf.AddV2"(%327, %344) {device = ""} : (tensor<?
↪x14x14x64xf16>, tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_9_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %346 = "tf.Conv2D"(%345, %13) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %351 = "tf.Relu6"(%350) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_10_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_10_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %352 = "tf.DepthwiseConv2dNative"(%351, %10) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_10_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %357 = "tf.Relu6"(%356) {device = ""} : (tensor<?x14x14x384xf16>)␣
↪-> tensor<?x14x14x384xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %358 = "tf.Conv2D"(%357, %16) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x96xf16>) -> tensor<?x14x14x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_10_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %363 = "tf.Conv2D"(%362, %22) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x96xf16>,␣
↪tensor<1x1x96x576xf16>) -> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %368 = "tf.Relu6"(%367) {device = ""} : (tensor<?x14x14x576xf16>)␣
↪-> tensor<?x14x14x576xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_11_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_11_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %369 = "tf.DepthwiseConv2dNative"(%368, %19) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<3x3x576x1xf16>) -> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_11_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %374 = "tf.Relu6"(%373) {device = ""} : (tensor<?x14x14x576xf16>)␣
↪-> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %375 = "tf.Conv2D"(%374, %25) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<1x1x576x96xf16>) -> tensor<?x14x14x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %380 = "tf.AddV2"(%362, %379) {device = ""} : (tensor<?
↪x14x14x96xf16>, tensor<?x14x14x96xf16>) -> tensor<?x14x14x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_11_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %381 = "tf.Conv2D"(%380, %31) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x96xf16>,␣
↪tensor<1x1x96x576xf16>) -> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
```

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<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %386 = "tf.Relu6"(%385) {device = ""} : (tensor<?x14x14x576xf16>)␣
↪-> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_12_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_12_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %387 = "tf.DepthwiseConv2dNative"(%386, %28) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<3x3x576x1xf16>) -> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_12_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %392 = "tf.Relu6"(%391) {device = ""} : (tensor<?x14x14x576xf16>)␣
↪-> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %393 = "tf.Conv2D"(%392, %34) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<1x1x576x96xf16>) -> tensor<?x14x14x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %398 = "tf.AddV2"(%380, %397) {device = ""} : (tensor<?
↪x14x14x96xf16>, tensor<?x14x14x96xf16>) -> tensor<?x14x14x96xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_12_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %399 = "tf.Conv2D"(%398, %40) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x96xf16>,␣
↪tensor<1x1x96x576xf16>) -> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %404 = "tf.Relu6"(%403) {device = ""} : (tensor<?x14x14x576xf16>)␣
↪-> tensor<?x14x14x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Pad' op
↪is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %405 = "tf.Pad"(%404, %5) {device = ""} : (tensor<?x14x14x576xf16>␣
↪tensor<4x2xi32>) -> tensor<?x15x15x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Pad:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_pad_1/Pad@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_13_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_13_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %406 = "tf.DepthwiseConv2dNative"(%405, %37) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "VALID",␣
↪strides = [1, 2, 2, 1]}> {device = ""} : (tensor<?x15x15x576xf16>,␣
↪tensor<3x3x576x1xf16>) -> tensor<?x7x7x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_13_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %411 = "tf.Relu6"(%410) {device = ""} : (tensor<?x7x7x576xf16>) ->␣
↪tensor<?x7x7x576xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
```

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<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %412 = "tf.Conv2D"(%411, %43) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x576xf16>,␣
↪tensor<1x1x576x160xf16>) -> tensor<?x7x7x160xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_13_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %417 = "tf.Conv2D"(%416, %49) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x160xf16>,␣
↪tensor<1x1x160x960xf16>) -> tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
```

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<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %422 = "tf.Relu6"(%421) {device = ""} : (tensor<?x7x7x960xf16>) ->
↪tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_14_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_14_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %423 = "tf.DepthwiseConv2dNative"(%422, %46) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<3x3x960x1xf16>) -> tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_14_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %428 = "tf.Relu6"(%427) {device = ""} : (tensor<?x7x7x960xf16>) ->
↪tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %429 = "tf.Conv2D"(%428, %52) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<1x1x960x160xf16>) -> tensor<?x7x7x160xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %434 = "tf.AddV2"(%416, %433) {device = ""} : (tensor<?
↪x7x7x160xf16>, tensor<?x7x7x160xf16>) -> tensor<?x7x7x160xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_14_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %435 = "tf.Conv2D"(%434, %58) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x160xf16>,␣
↪tensor<1x1x160x960xf16>) -> tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %440 = "tf.Relu6"(%439) {device = ""} : (tensor<?x7x7x960xf16>) -> ␣
↪tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_15_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_15_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %441 = "tf.DepthwiseConv2dNative"(%440, %55) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<3x3x960x1xf16>) -> tensor<?x7x7x960xf16>
```

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<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_15_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %446 = "tf.Relu6"(%445) {device = ""} : (tensor<?x7x7x960xf16>) -> ␣
↪tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %447 = "tf.Conv2D"(%446, %61) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<1x1x960x160xf16>) -> tensor<?x7x7x160xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.AddV2'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
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<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %452 = "tf.AddV2"(%434, %451) {device = ""} : (tensor<?
↪x7x7x160xf16>, tensor<?x7x7x160xf16>) -> tensor<?x7x7x160xf16>
<unknown>:0: note: loc(callsite(callsite(fused["AddV2:", "functional_1/
↪mobilenetv2_1.00_224_1/block_15_add_1/Add@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %453 = "tf.Conv2D"(%452, %67) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x160xf16>,␣
↪tensor<1x1x160x960xf16>) -> tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_expand_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current
↪operation: %458 = "tf.Relu6"(%457) {device = ""} : (tensor<?x7x7x960xf16>) ->
↪tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_expand_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_16_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.
↪DepthwiseConv2dNative' op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_16_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %459 = "tf.DepthwiseConv2dNative"(%458, %64) <{data_format =␣
↪"NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME",␣
↪strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<3x3x960x1xf16>) -> tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["DepthwiseConv2dNative:",␣
↪"functional_1/mobilenetv2_1.00_224_1/block_16_depthwise_1/
↪depthwise@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %464 = "tf.Relu6"(%463) {device = ""} : (tensor<?x7x7x960xf16>) ->␣
↪tensor<?x7x7x960xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_depthwise_relu_1/
↪Relu6@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
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<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %465 = "tf.Conv2D"(%464, %70) <{data_format = "NHWC", dilations =␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<1x1x960x320xf16>) -> tensor<?x7x7x320xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/block_16_project_1/
↪convolution@__inference_function_208818"] at fused["StatefulPartitionedCall:" ␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv_1_1/convolution@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Conv2D' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv_1_1/convolution@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %470 = "tf.Conv2D"(%469, %157) <{data_format = "NHWC", dilations = ␣
↪[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 1, 1,␣
↪1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x320xf16>,␣
↪tensor<1x1x320x1280xf16>) -> tensor<?x7x7x1280xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Conv2D:", "functional_1/
↪mobilenetv2_1.00_224_1/Conv_1_1/convolution@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/out_relu_1/Relu6@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu6'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/out_relu_1/Relu6@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %475 = "tf.Relu6"(%474) {device = ""} : (tensor<?x7x7x1280xf16>) -␣
↪tensor<?x7x7x1280xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu6:", "functional_1/
↪mobilenetv2_1.00_224_1/out_relu_1/Relu6@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["MatMul:", "functional_1/dense_ /
↪MatMul@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.MatMul' ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["MatMul:", "functional_1/dense_1
↪MatMul@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %483 = "tf.MatMul"(%482, %2) <{grad_a = false, grad_b = false,␣
↪transpose_a = false, transpose_b = true}> : (tensor<?x1280xf16>,␣
↪tensor<128x1280xf16>) -> tensor<?x128xf16>
<unknown>:0: note: loc(callsite(callsite(fused["MatMul:", "functional_1/dense_1
↪MatMul@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["BiasAdd:", "functional_1/
↪dense_1/BiasAdd@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.BiasAdd ␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["BiasAdd:", "functional_1/dense_ /
↪BiasAdd@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %484 = "tf.BiasAdd"(%483, %9) <{data_format = "NHWC"}> {device =␣
↪""} : (tensor<?x128xf16>, tensor<128xf16>) -> tensor<?x128xf16>
<unknown>:0: note: loc(callsite(callsite(fused["BiasAdd:", "functional_1/dense_ /
↪BiasAdd@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu:", "functional_1/dense_1/
↪Relu@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu' op␣
↪is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu:", "functional_1/dense_1/
↪Relu@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current ␣
↪operation: %485 = "tf.Relu"(%484) {device = ""} : (tensor<?x128xf16>) ->␣
↪tensor<?x128xf16>
<unknown>:0: note: loc(callsite(callsite(fused["Relu:", "functional_1/dense_1/
↪Relu@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code: ␣
↪ERROR_NEEDS_FLEX_OPS
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<unknown>:0: error: loc(callsite(callsite(fused["MatMul:", "functional_1/
↪dense_1_2/MatMul@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.MatMul'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["MatMul:", "functional_1/
↪dense_1_2/MatMul@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %486 = "tf.MatMul"(%485, %1) <{grad_a = false, grad_b = false,␣
↪transpose_a = false, transpose_b = true}> : (tensor<?x128xf16>,␣
↪tensor<64x128xf16>) -> tensor<?x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["MatMul:", "functional_1/
↪dense_1_2/MatMul@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["BiasAdd:", "functional_1/
↪dense_1_2/BiasAdd@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.BiasAdd'␣
↪op is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["BiasAdd:", "functional_1/
↪dense_1_2/BiasAdd@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %487 = "tf.BiasAdd"(%486, %8) <{data_format = "NHWC"}> {device =␣
↪""} : (tensor<?x64xf16>, tensor<64xf16>) -> tensor<?x64xf16>
<unknown>:0: note: loc(callsite(callsite(fused["BiasAdd:", "functional_1/
↪dense_1_2/BiasAdd@__inference_function_208818"] at␣
↪fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: loc(callsite(callsite(fused["Relu:", "functional_1/dense_1_2/
↪Relu@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): 'tf.Relu' op␣
↪is neither a custom op nor a flex op
<unknown>:0: note: loc(fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall_1"]): called from
<unknown>:0: note: loc(callsite(callsite(fused["Relu:", "functional_1/dense_1_2/
↪Relu@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): see current␣
↪operation: %488 = "tf.Relu"(%487) {device = ""} : (tensor<?x64xf16>) ->␣
↪tensor<?x64xf16>
```

```
<unknown>:0: note: loc(callsite(callsite(fused["Relu:", "functional_1/dense_1_2
↪Relu@__inference_function_208818"] at fused["StatefulPartitionedCall:",␣
↪"StatefulPartitionedCall@__inference_signature_wrapper_209909"]) at␣
↪fused["StatefulPartitionedCall:", "StatefulPartitionedCall_1"])): Error code:␣
↪ERROR_NEEDS_FLEX_OPS
<unknown>:0: error: failed while converting: 'main':
Some ops are not supported by the native TFLite runtime, you can enable TF␣
↪kernels fallback using TF Select. See instructions: https://www.tensorflow.or;/
↪lite/guide/ops_select
TF Select ops: AddV2, BiasAdd, Conv2D, DepthwiseConv2dNative, MatMul, Pad, Relu␣
↪Relu6
Details:
        tf.AddV2(tensor<?x14x14x64xf16>, tensor<?x14x14x64xf16>) -> (tensor<?
↪x14x14x64xf16>) : {device = ""}
        tf.AddV2(tensor<?x14x14x96xf16>, tensor<?x14x14x96xf16>) -> (tensor<?
↪x14x14x96xf16>) : {device = ""}
        tf.AddV2(tensor<?x28x28x32xf16>, tensor<?x28x28x32xf16>) -> (tensor<?
↪x28x28x32xf16>) : {device = ""}
        tf.AddV2(tensor<?x56x56x24xf16>, tensor<?x56x56x24xf16>) -> (tensor<?
↪x56x56x24xf16>) : {device = ""}
        tf.AddV2(tensor<?x7x7x160xf16>, tensor<?x7x7x160xf16>) -> (tensor<?
↪x7x7x160xf16>) : {device = ""}
        tf.BiasAdd(tensor<?x128xf16>, tensor<128xf16>) -> (tensor<?x128xf16>) :␣
↪{data_format = "NHWC", device = ""}
        tf.BiasAdd(tensor<?x64xf16>, tensor<64xf16>) -> (tensor<?x64xf16>) :␣
↪{data_format = "NHWC", device = ""}
        tf.Conv2D(tensor<?x112x112x16xf16>, tensor<1x1x16x96xf16>) -> (tensor<?
↪x112x112x96xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x112x112x32xf16>, tensor<1x1x32x16xf16>) -> (tensor<?
↪x112x112x16xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x192xf16>, tensor<1x1x192x64xf16>) -> (tensor<?
↪x14x14x64xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x384xf16>, tensor<1x1x384x64xf16>) -> (tensor<?
↪x14x14x64xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x384xf16>, tensor<1x1x384x96xf16>) -> (tensor<?
↪x14x14x96xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
        tf.Conv2D(tensor<?x14x14x576xf16>, tensor<1x1x576x96xf16>) -> (tensor<?
↪x14x14x96xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
```

```
      tf.Conv2D(tensor<?x14x14x64xf16>, tensor<1x1x64x384xf16>) -> (tensor<?
↪x14x14x384xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x14x14x96xf16>, tensor<1x1x96x576xf16>) -> (tensor<?
↪x14x14x576xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x224x224x3xf16>, tensor<3x3x3x32xf16>) -> (tensor<?
↪x112x112x32xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 2, 2, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x28x28x144xf16>, tensor<1x1x144x32xf16>) -> (tensor<?
↪x28x28x32xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x28x28x192xf16>, tensor<1x1x192x32xf16>) -> (tensor<?
↪x28x28x32xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x28x28x32xf16>, tensor<1x1x32x192xf16>) -> (tensor<?
↪x28x28x192xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x56x56x144xf16>, tensor<1x1x144x24xf16>) -> (tensor<?
↪x56x56x24xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x56x56x24xf16>, tensor<1x1x24x144xf16>) -> (tensor<?
↪x56x56x144xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x56x56x96xf16>, tensor<1x1x96x24xf16>) -> (tensor<?
↪x56x56x24xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x7x7x160xf16>, tensor<1x1x160x960xf16>) -> (tensor<?
↪x7x7x960xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1, 1] ␣
↪explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x7x7x320xf16>, tensor<1x1x320x1280xf16>) -> (tensor<?
↪x7x7x1280xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "VALID", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x7x7x576xf16>, tensor<1x1x576x160xf16>) -> (tensor<?
↪x7x7x160xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1, 1] ␣
↪explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x7x7x960xf16>, tensor<1x1x960x160xf16>) -> (tensor<?
↪x7x7x160xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1, 1] ␣
↪explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
      tf.Conv2D(tensor<?x7x7x960xf16>, tensor<1x1x960x320xf16>) -> (tensor<?
↪x7x7x320xf16>) : {data_format = "NHWC", device = "", dilations = [1, 1, 1, 1] ␣
↪explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}
```

```
      tf.DepthwiseConv2dNative(tensor<?x112x112x32xf16>, tensor<3x3x32x1xf16>
↪-> (tensor<?x112x112x32xf16>) : {data_format = "NHWC", device = "", dilations
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,
↪1]}
      tf.DepthwiseConv2dNative(tensor<?x113x113x96xf16>, tensor<3x3x96x1xf16>
↪-> (tensor<?x56x56x96xf16>) : {data_format = "NHWC", device = "", dilations =
↪[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2,
↪1]}
      tf.DepthwiseConv2dNative(tensor<?x14x14x384xf16>, tensor<3x3x384x1xf16>
↪-> (tensor<?x14x14x384xf16>) : {data_format = "NHWC", device = "", dilations
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1 }
      tf.DepthwiseConv2dNative(tensor<?x14x14x576xf16>, tensor<3x3x576x1xf16>
↪-> (tensor<?x14x14x576xf16>) : {data_format = "NHWC", device = "", dilations
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1 }
      tf.DepthwiseConv2dNative(tensor<?x15x15x576xf16>, tensor<3x3x576x1xf16>
↪-> (tensor<?x7x7x576xf16>) : {data_format = "NHWC", device = "", dilations =
↪[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2,
↪1]}
      tf.DepthwiseConv2dNative(tensor<?x28x28x192xf16>, tensor<3x3x192x1xf16>
↪-> (tensor<?x28x28x192xf16>) : {data_format = "NHWC", device = "", dilations
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1 }
      tf.DepthwiseConv2dNative(tensor<?x29x29x192xf16>, tensor<3x3x192x1xf16>
↪-> (tensor<?x14x14x192xf16>) : {data_format = "NHWC", device = "", dilations
↪[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2,
↪1]}
      tf.DepthwiseConv2dNative(tensor<?x56x56x144xf16>, tensor<3x3x144x1xf16>
↪-> (tensor<?x56x56x144xf16>) : {data_format = "NHWC", device = "", dilations
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1 }
      tf.DepthwiseConv2dNative(tensor<?x57x57x144xf16>, tensor<3x3x144x1xf16>
↪-> (tensor<?x28x28x144xf16>) : {data_format = "NHWC", device = "", dilations
↪[1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2,
↪1]}
      tf.DepthwiseConv2dNative(tensor<?x7x7x960xf16>, tensor<3x3x960x1xf16>)
↪-> (tensor<?x7x7x960xf16>) : {data_format = "NHWC", device = "", dilations =
↪[1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1 }
      tf.MatMul(tensor<?x1280xf16>, tensor<128x1280xf16>) -> (tensor<?
↪x128xf16>) : {grad_a = false, grad_b = false, transpose_a = false, transpose_
↪= true}
      tf.MatMul(tensor<?x128xf16>, tensor<64x128xf16>) -> (tensor<?x64xf16>)
↪{grad_a = false, grad_b = false, transpose_a = false, transpose_b = true}
      tf.Pad(tensor<?x112x112x96xf16>, tensor<4x2xi32>) -> (tensor<?
↪x113x113x96xf16>) : {device = ""}
      tf.Pad(tensor<?x14x14x576xf16>, tensor<4x2xi32>) -> (tensor<?
↪x15x15x576xf16>) : {device = ""}
      tf.Pad(tensor<?x28x28x192xf16>, tensor<4x2xi32>) -> (tensor<?
↪x29x29x192xf16>) : {device = ""}
      tf.Pad(tensor<?x56x56x144xf16>, tensor<4x2xi32>) -> (tensor<?
↪x57x57x144xf16>) : {device = ""}
      tf.Relu(tensor<?x128xf16>) -> (tensor<?x128xf16>) : {device = ""}
      tf.Relu(tensor<?x64xf16>) -> (tensor<?x64xf16>) : {device = ""}
      tf.Relu6(tensor<?x112x112x32xf16>) -> (tensor<?x112x112x32xf16>) :
↪{device = ""}
```

```
        tf.Relu6(tensor<?x112x112x96xf16>) -> (tensor<?x112x112x96xf16>) :␣
↪{device = ""}
        tf.Relu6(tensor<?x14x14x192xf16>) -> (tensor<?x14x14x192xf16>) : {devic␣
↪= ""}
        tf.Relu6(tensor<?x14x14x384xf16>) -> (tensor<?x14x14x384xf16>) : {devic␣
↪= ""}
        tf.Relu6(tensor<?x14x14x576xf16>) -> (tensor<?x14x14x576xf16>) : {devic␣
↪= ""}
        tf.Relu6(tensor<?x28x28x144xf16>) -> (tensor<?x28x28x144xf16>) : {devic␣
↪= ""}
        tf.Relu6(tensor<?x28x28x192xf16>) -> (tensor<?x28x28x192xf16>) : {devic␣
↪= ""}
        tf.Relu6(tensor<?x56x56x144xf16>) -> (tensor<?x56x56x144xf16>) : {devic␣
↪= ""}
        tf.Relu6(tensor<?x56x56x96xf16>) -> (tensor<?x56x56x96xf16>) : {device =␣
↪""}
        tf.Relu6(tensor<?x7x7x1280xf16>) -> (tensor<?x7x7x1280xf16>) : {device =␣
↪""}
        tf.Relu6(tensor<?x7x7x576xf16>) -> (tensor<?x7x7x576xf16>) : {device =␣
↪""}
        tf.Relu6(tensor<?x7x7x960xf16>) -> (tensor<?x7x7x960xf16>) : {device =␣
↪""}

<unknown>:0: note: see current operation:
"func.func"() <{arg_attrs = [{tf_saved_model.index_path = ["input_layer_1"]}],␣
↪function_type = (tensor<?x224x224x3xf32>) -> tensor<?x4xf32>, res_attrs =␣
↪[{tf_saved_model.index_path = ["output_0"]}], sym_name = "main"}> ({
^bb0(%arg0: tensor<?x224x224x3xf32>):
  %0 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<4x64xf32>}> : () -> tensor<4x64xf32>
  %1 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64x128xf16>}> : () -> tensor<64x128xf16>
  %2 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<128x1280xf16>}> : () -> tensor<128x1280xf16>
  %3 = "arith.constant"() <{value = dense<[-0.00107014622, 1.800760e-02, -0.
↪00126225909, -0.0169418678]> : tensor<4xf32>}> : () -> tensor<4xf32>
  %4 = "arith.constant"() <{value = dense<[1, 2]> : tensor<2xi32>}> : () ->␣
↪tensor<2xi32>
  %5 = "arith.constant"() <{value = dense<[[0, 0], [0, 1], [0, 1], [0, 0]]> :␣
↪tensor<4x2xi32>}> : () -> tensor<4x2xi32>
  %6 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1280xf32>}> : () -> tensor<1280xf32>
  %7 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1280xf32>}> : () -> tensor<1280xf32>
  %8 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf16>}> : () -> tensor<64xf16>
  %9 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<128xf16>}> : () -> tensor<128xf16>
```

```
%10 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x384x1xf16>}> : () -> tensor<3x3x384x1xf16>
%11 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%12 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%13 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x64x384xf16>}> : () -> tensor<1x1x64x384xf16>
%14 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%15 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%16 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x384x96xf16>}> : () -> tensor<1x1x384x96xf16>
%17 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%18 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%19 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x576x1xf16>}> : () -> tensor<3x3x576x1xf16>
%20 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%21 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%22 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x96x576xf16>}> : () -> tensor<1x1x96x576xf16>
%23 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%24 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%25 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x576x96xf16>}> : () -> tensor<1x1x576x96xf16>
%26 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%27 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%28 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x576x1xf16>}> : () -> tensor<3x3x576x1xf16>
%29 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%30 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%31 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x96x576xf16>}> : () -> tensor<1x1x96x576xf16>
%32 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%33 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
```

```
%34 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x576x96xf16>}> : () -> tensor<1x1x576x96xf16>
%35 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%36 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%37 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x576x1xf16>}> : () -> tensor<3x3x576x1xf16>
%38 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%39 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%40 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x96x576xf16>}> : () -> tensor<1x1x96x576xf16>
%41 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%42 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<576xf32>}> : () -> tensor<576xf32>
%43 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x576x160xf16>}> : () -> tensor<1x1x576x160xf16>
%44 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<160xf32>}> : () -> tensor<160xf32>
%45 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<160xf32>}> : () -> tensor<160xf32>
%46 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x960x1xf16>}> : () -> tensor<3x3x960x1xf16>
%47 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%48 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%49 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x160x960xf16>}> : () -> tensor<1x1x160x960xf16>
%50 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%51 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%52 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x960x160xf16>}> : () -> tensor<1x1x960x160xf16>
%53 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<160xf32>}> : () -> tensor<160xf32>
%54 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<160xf32>}> : () -> tensor<160xf32>
%55 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x960x1xf16>}> : () -> tensor<3x3x960x1xf16>
%56 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%57 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
```

```
%58 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x160x960xf16>}> : () -> tensor<1x1x160x960xf16>
%59 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%60 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%61 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x960x160xf16>}> : () -> tensor<1x1x960x160xf16>
%62 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<160xf32>}> : () -> tensor<160xf32>
%63 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<160xf32>}> : () -> tensor<160xf32>
%64 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x960x1xf16>}> : () -> tensor<3x3x960x1xf16>
%65 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%66 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%67 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x160x960xf16>}> : () -> tensor<1x1x160x960xf16>
%68 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%69 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<960xf32>}> : () -> tensor<960xf32>
%70 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x960x320xf16>}> : () -> tensor<1x1x960x320xf16>
%71 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<320xf32>}> : () -> tensor<320xf32>
%72 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<320xf32>}> : () -> tensor<320xf32>
%73 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x96x1xf16>}> : () -> tensor<3x3x96x1xf16>
%74 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%75 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%76 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x16x96xf16>}> : () -> tensor<1x1x16x96xf16>
%77 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%78 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<96xf32>}> : () -> tensor<96xf32>
%79 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x96x24xf16>}> : () -> tensor<1x1x96x24xf16>
%80 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<24xf32>}> : () -> tensor<24xf32>
%81 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<24xf32>}> : () -> tensor<24xf32>
```

```
%82 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x144x1xf16>}> : () -> tensor<3x3x144x1xf16>
%83 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%84 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%85 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x24x144xf16>}> : () -> tensor<1x1x24x144xf16>
%86 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%87 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%88 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x144x24xf16>}> : () -> tensor<1x1x144x24xf16>
%89 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<24xf32>}> : () -> tensor<24xf32>
%90 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<24xf32>}> : () -> tensor<24xf32>
%91 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x144x1xf16>}> : () -> tensor<3x3x144x1xf16>
%92 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%93 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%94 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x24x144xf16>}> : () -> tensor<1x1x24x144xf16>
%95 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%96 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<144xf32>}> : () -> tensor<144xf32>
%97 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x144x32xf16>}> : () -> tensor<1x1x144x32xf16>
%98 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<32xf32>}> : () -> tensor<32xf32>
%99 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<32xf32>}> : () -> tensor<32xf32>
%100 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x192x1xf16>}> : () -> tensor<3x3x192x1xf16>
%101 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%102 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%103 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x32x192xf16>}> : () -> tensor<1x1x32x192xf16>
%104 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%105 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
```

```
%106 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x192x32xf16>}> : () -> tensor<1x1x192x32xf16>
%107 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<32xf32>}> : () -> tensor<32xf32>
%108 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<32xf32>}> : () -> tensor<32xf32>
%109 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x192x1xf16>}> : () -> tensor<3x3x192x1xf16>
%110 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%111 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%112 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x32x192xf16>}> : () -> tensor<1x1x32x192xf16>
%113 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%114 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%115 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x192x32xf16>}> : () -> tensor<1x1x192x32xf16>
%116 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<32xf32>}> : () -> tensor<32xf32>
%117 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<32xf32>}> : () -> tensor<32xf32>
%118 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x192x1xf16>}> : () -> tensor<3x3x192x1xf16>
%119 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%120 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%121 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x32x192xf16>}> : () -> tensor<1x1x32x192xf16>
%122 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%123 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<192xf32>}> : () -> tensor<192xf32>
%124 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x192x64xf16>}> : () -> tensor<1x1x192x64xf16>
%125 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%126 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%127 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x384x1xf16>}> : () -> tensor<3x3x384x1xf16>
%128 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%129 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
```

```
%130 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x64x384xf16>}> : () -> tensor<1x1x64x384xf16>
%131 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%132 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%133 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x384x64xf16>}> : () -> tensor<1x1x384x64xf16>
%134 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%135 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%136 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x384x1xf16>}> : () -> tensor<3x3x384x1xf16>
%137 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%138 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%139 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x64x384xf16>}> : () -> tensor<1x1x64x384xf16>
%140 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%141 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%142 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x384x64xf16>}> : () -> tensor<1x1x384x64xf16>
%143 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%144 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%145 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<3x3x384x1xf16>}> : () -> tensor<3x3x384x1xf16>
%146 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%147 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%148 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x64x384xf16>}> : () -> tensor<1x1x64x384xf16>
%149 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%150 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<384xf32>}> : () -> tensor<384xf32>
%151 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<1x1x384x64xf16>}> : () -> tensor<1x1x384x64xf16>
%152 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
%153 = "arith.constant"() <{value = dense_resource<__elided__> :␣
↪tensor<64xf32>}> : () -> tensor<64xf32>
```

```
%154 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<32xf32>}> : () -> tensor<32xf32>
%155 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<32xf32>}> : () -> tensor<32xf32>
%156 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<3x3x3x32xf16>}> : () -> tensor<3x3x3x32xf16>
%157 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<1x1x320x1280xf16>}> : () -> tensor<1x1x320x1280xf16>
%158 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<1280xf32>}> : () -> tensor<1280xf32>
%159 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<1280xf32>}> : () -> tensor<1280xf32>
%160 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<3x3x32x1xf16>}> : () -> tensor<3x3x32x1xf16>
%161 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<32xf32>}> : () -> tensor<32xf32>
%162 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<32xf32>}> : () -> tensor<32xf32>
%163 = "arith.constant"() <{value = dense_resource<__elided__> : tensor<1x1x32x16xf16>}> : () -> tensor<1x1x32x16xf16>
%164 = "arith.constant"() <{value = dense<[4.83980179, 7.01497889, 6.59778308, 7.4643445, 3.82265639, 6.78915548, 5.1570487, 4.63491488, 4.46355867, 6.7842741, 7.31219244, 4.33215332, 6.80388832, 6.5691595, 5.91440105, 4.81517363]> : tensor<16xf32>}> : () -> tensor<16xf32>
%165 = "arith.constant"() <{value = dense<[-1.79617035, 3.17504382, 16.2980614, 24.0359573, 13.8513613, 3.49967337, -0.0168914795, 7.31451178, -13.8333397, 3.81765318, -2.79449153, 18.9408112, -18.930912, -11.2135153, 7.71239185, -2.86968422]> : tensor<16xf32>}> : () -> tensor<16xf32>
%166 = "tfl.cast"(%arg0) : (tensor<?x224x224x3xf32>) -> tensor<?x224x224x3xf1
%167 = "tf.Conv2D"(%166, %156) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 2, 2, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x224x224x3xf16>, tensor<3x3x3x32xf16>) -> tensor<?x112x112x32xf16>
%168 = "tfl.cast"(%167) : (tensor<?x112x112x32xf16>) -> tensor<?x112x112x32xf32>
%169 = "tfl.mul"(%168, %154) <{fused_activation_function = "NONE"}> : (tensor ?x112x112x32xf32>, tensor<32xf32>) -> tensor<?x112x112x32xf32>
%170 = "tfl.add"(%169, %155) <{fused_activation_function = "NONE"}> : (tensor ?x112x112x32xf32>, tensor<32xf32>) -> tensor<?x112x112x32xf32>
%171 = "tfl.cast"(%170) : (tensor<?x112x112x32xf32>) -> tensor<?x112x112x32xf16>
%172 = "tf.Relu6"(%171) {device = ""} : (tensor<?x112x112x32xf16>) -> tensor<x112x112x32xf16>
%173 = "tf.DepthwiseConv2dNative"(%172, %160) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x112x112x32xf16>, tensor<3x3x32x1xf16>) -> tensor<?x112x112x32xf16>
%174 = "tfl.cast"(%173) : (tensor<?x112x112x32xf16>) -> tensor<?x112x112x32xf32>
```

```
%175 = "tfl.mul"(%174, %161) <{fused_activation_function = "NONE"}> : (tensor<?x112x112x32xf32>, tensor<32xf32>) -> tensor<?x112x112x32xf32>
%176 = "tfl.add"(%175, %162) <{fused_activation_function = "NONE"}> : (tensor<?x112x112x32xf32>, tensor<32xf32>) -> tensor<?x112x112x32xf32>
%177 = "tfl.cast"(%176) : (tensor<?x112x112x32xf32>) -> tensor<?x112x112x32xf16>
%178 = "tf.Relu6"(%177) {device = ""} : (tensor<?x112x112x32xf16>) -> tensor<?x112x112x32xf16>
%179 = "tf.Conv2D"(%178, %163) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x112x112x32xf16>, tensor<1x1x32x16xf16>) -> tensor<?x112x112x16xf16>
%180 = "tfl.cast"(%179) : (tensor<?x112x112x16xf16>) -> tensor<?x112x112x16xf32>
%181 = "tfl.mul"(%180, %164) <{fused_activation_function = "NONE"}> : (tensor<?x112x112x16xf32>, tensor<16xf32>) -> tensor<?x112x112x16xf32>
%182 = "tfl.add"(%181, %165) <{fused_activation_function = "NONE"}> : (tensor<?x112x112x16xf32>, tensor<16xf32>) -> tensor<?x112x112x16xf32>
%183 = "tfl.cast"(%182) : (tensor<?x112x112x16xf32>) -> tensor<?x112x112x16xf16>
%184 = "tf.Conv2D"(%183, %76) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x112x112x16xf16>, tensor<1x1x16x96xf16>) -> tensor<?x112x112x96xf16>
%185 = "tfl.cast"(%184) : (tensor<?x112x112x96xf16>) -> tensor<?x112x112x96xf32>
%186 = "tfl.mul"(%185, %77) <{fused_activation_function = "NONE"}> : (tensor<?x112x112x96xf32>, tensor<96xf32>) -> tensor<?x112x112x96xf32>
%187 = "tfl.add"(%186, %78) <{fused_activation_function = "NONE"}> : (tensor<?x112x112x96xf32>, tensor<96xf32>) -> tensor<?x112x112x96xf32>
%188 = "tfl.cast"(%187) : (tensor<?x112x112x96xf32>) -> tensor<?x112x112x96xf16>
%189 = "tf.Relu6"(%188) {device = ""} : (tensor<?x112x112x96xf16>) -> tensor<?x112x112x96xf16>
%190 = "tf.Pad"(%189, %5) {device = ""} : (tensor<?x112x112x96xf16>, tensor<4x2xi32>) -> tensor<?x113x113x96xf16>
%191 = "tf.DepthwiseConv2dNative"(%190, %73) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2, 1]}> {device = ""} : (tensor<?x113x113x96xf16>, tensor<3x3x96x1xf16>) -> tensor<?x56x56x96xf16>
%192 = "tfl.cast"(%191) : (tensor<?x56x56x96xf16>) -> tensor<?x56x56x96xf32>
%193 = "tfl.mul"(%192, %74) <{fused_activation_function = "NONE"}> : (tensor<?x56x56x96xf32>, tensor<96xf32>) -> tensor<?x56x56x96xf32>
%194 = "tfl.add"(%193, %75) <{fused_activation_function = "NONE"}> : (tensor<?x56x56x96xf32>, tensor<96xf32>) -> tensor<?x56x56x96xf32>
%195 = "tfl.cast"(%194) : (tensor<?x56x56x96xf32>) -> tensor<?x56x56x96xf16>
%196 = "tf.Relu6"(%195) {device = ""} : (tensor<?x56x56x96xf16>) -> tensor<?x56x56x96xf16>
```

```
%197 = "tf.Conv2D"(%196, %79) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x96xf16>,␣
↪tensor<1x1x96x24xf16>) -> tensor<?x56x56x24xf16>
%198 = "tfl.cast"(%197) : (tensor<?x56x56x24xf16>) -> tensor<?x56x56x24xf32>
%199 = "tfl.mul"(%198, %80) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x24xf32>, tensor<24xf32>) -> tensor<?x56x56x24xf32>
%200 = "tfl.add"(%199, %81) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x24xf32>, tensor<24xf32>) -> tensor<?x56x56x24xf32>
%201 = "tfl.cast"(%200) : (tensor<?x56x56x24xf32>) -> tensor<?x56x56x24xf16>
%202 = "tf.Conv2D"(%201, %85) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x24xf16>,␣
↪tensor<1x1x24x144xf16>) -> tensor<?x56x56x144xf16>
%203 = "tfl.cast"(%202) : (tensor<?x56x56x144xf16>) -> tensor<?x56x56x144xf32
%204 = "tfl.mul"(%203, %86) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x144xf32>, tensor<144xf32>) -> tensor<?x56x56x144xf32>
%205 = "tfl.add"(%204, %87) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x144xf32>, tensor<144xf32>) -> tensor<?x56x56x144xf32>
%206 = "tfl.cast"(%205) : (tensor<?x56x56x144xf32>) -> tensor<?x56x56x144xf16
%207 = "tf.Relu6"(%206) {device = ""} : (tensor<?x56x56x144xf16>) -> tensor<?
↪x56x56x144xf16>
%208 = "tf.DepthwiseConv2dNative"(%207, %82) <{data_format = "NHWC", dilation␣
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1]}> {device = ""} : (tensor<?x56x56x144xf16>, tensor<3x3x144x1xf16>) ->␣
↪tensor<?x56x56x144xf16>
%209 = "tfl.cast"(%208) : (tensor<?x56x56x144xf16>) -> tensor<?x56x56x144xf32
%210 = "tfl.mul"(%209, %83) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x144xf32>, tensor<144xf32>) -> tensor<?x56x56x144xf32>
%211 = "tfl.add"(%210, %84) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x144xf32>, tensor<144xf32>) -> tensor<?x56x56x144xf32>
%212 = "tfl.cast"(%211) : (tensor<?x56x56x144xf32>) -> tensor<?x56x56x144xf16
%213 = "tf.Relu6"(%212) {device = ""} : (tensor<?x56x56x144xf16>) -> tensor<?
↪x56x56x144xf16>
%214 = "tf.Conv2D"(%213, %88) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x144xf16>,␣
↪tensor<1x1x144x24xf16>) -> tensor<?x56x56x24xf16>
%215 = "tfl.cast"(%214) : (tensor<?x56x56x24xf16>) -> tensor<?x56x56x24xf32>
%216 = "tfl.mul"(%215, %89) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x24xf32>, tensor<24xf32>) -> tensor<?x56x56x24xf32>
%217 = "tfl.add"(%216, %90) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x24xf32>, tensor<24xf32>) -> tensor<?x56x56x24xf32>
%218 = "tfl.cast"(%217) : (tensor<?x56x56x24xf32>) -> tensor<?x56x56x24xf16>
%219 = "tf.AddV2"(%201, %218) {device = ""} : (tensor<?x56x56x24xf16>, tensor ?
↪x56x56x24xf16>) -> tensor<?x56x56x24xf16>
%220 = "tf.Conv2D"(%219, %94) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x56x56x24xf16>,␣
↪tensor<1x1x24x144xf16>) -> tensor<?x56x56x144xf16>
```

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%221 = "tfl.cast"(%220) : (tensor<?x56x56x144xf16>) -> tensor<?x56x56x144xf32
%222 = "tfl.mul"(%221, %95) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x144xf32>, tensor<144xf32>) -> tensor<?x56x56x144xf32>
%223 = "tfl.add"(%222, %96) <{fused_activation_function = "NONE"}> : (tensor<
↪x56x56x144xf32>, tensor<144xf32>) -> tensor<?x56x56x144xf32>
%224 = "tfl.cast"(%223) : (tensor<?x56x56x144xf32>) -> tensor<?x56x56x144xf16
%225 = "tf.Relu6"(%224) {device = ""} : (tensor<?x56x56x144xf16>) -> tensor<?
↪x56x56x144xf16>
%226 = "tf.Pad"(%225, %5) {device = ""} : (tensor<?x56x56x144xf16>,␣
↪tensor<4x2xi32>) -> tensor<?x57x57x144xf16>
%227 = "tf.DepthwiseConv2dNative"(%226, %91) <{data_format = "NHWC", dilation␣
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2 ␣
↪1]}> {device = ""} : (tensor<?x57x57x144xf16>, tensor<3x3x144x1xf16>) ->␣
↪tensor<?x28x28x144xf16>
%228 = "tfl.cast"(%227) : (tensor<?x28x28x144xf16>) -> tensor<?x28x28x144xf32
%229 = "tfl.mul"(%228, %92) <{fused_activation_function = "NONE"}> : (tensor<
↪x28x28x144xf32>, tensor<144xf32>) -> tensor<?x28x28x144xf32>
%230 = "tfl.add"(%229, %93) <{fused_activation_function = "NONE"}> : (tensor<
↪x28x28x144xf32>, tensor<144xf32>) -> tensor<?x28x28x144xf32>
%231 = "tfl.cast"(%230) : (tensor<?x28x28x144xf32>) -> tensor<?x28x28x144xf16
%232 = "tf.Relu6"(%231) {device = ""} : (tensor<?x28x28x144xf16>) -> tensor<?
↪x28x28x144xf16>
%233 = "tf.Conv2D"(%232, %97) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x144xf16>,␣
↪tensor<1x1x144x32xf16>) -> tensor<?x28x28x32xf16>
%234 = "tfl.cast"(%233) : (tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf32>
%235 = "tfl.mul"(%234, %98) <{fused_activation_function = "NONE"}> : (tensor<
↪x28x28x32xf32>, tensor<32xf32>) -> tensor<?x28x28x32xf32>
%236 = "tfl.add"(%235, %99) <{fused_activation_function = "NONE"}> : (tensor<
↪x28x28x32xf32>, tensor<32xf32>) -> tensor<?x28x28x32xf32>
%237 = "tfl.cast"(%236) : (tensor<?x28x28x32xf32>) -> tensor<?x28x28x32xf16>
%238 = "tf.Conv2D"(%237, %103) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x32xf16>,␣
↪tensor<1x1x32x192xf16>) -> tensor<?x28x28x192xf16>
%239 = "tfl.cast"(%238) : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf32
%240 = "tfl.mul"(%239, %104) <{fused_activation_function = "NONE"}> : (tensor ?
↪x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%241 = "tfl.add"(%240, %105) <{fused_activation_function = "NONE"}> : (tensor ?
↪x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%242 = "tfl.cast"(%241) : (tensor<?x28x28x192xf32>) -> tensor<?x28x28x192xf16
%243 = "tf.Relu6"(%242) {device = ""} : (tensor<?x28x28x192xf16>) -> tensor<?
↪x28x28x192xf16>
%244 = "tf.DepthwiseConv2dNative"(%243, %100) <{data_format = "NHWC",␣
↪dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = ␣
↪[1, 1, 1, 1]}> {device = ""} : (tensor<?x28x28x192xf16>,␣
↪tensor<3x3x192x1xf16>) -> tensor<?x28x28x192xf16>
%245 = "tfl.cast"(%244) : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf32
```

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%246 = "tfl.mul"(%245, %101) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%247 = "tfl.add"(%246, %102) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%248 = "tfl.cast"(%247) : (tensor<?x28x28x192xf32>) -> tensor<?x28x28x192xf16
%249 = "tf.Relu6"(%248) {device = ""} : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf16>
%250 = "tf.Conv2D"(%249, %106) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x192xf16>, tensor<1x1x192x32xf16>) -> tensor<?x28x28x32xf16>
%251 = "tfl.cast"(%250) : (tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf32>
%252 = "tfl.mul"(%251, %107) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x32xf32>, tensor<32xf32>) -> tensor<?x28x28x32xf32>
%253 = "tfl.add"(%252, %108) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x32xf32>, tensor<32xf32>) -> tensor<?x28x28x32xf32>
%254 = "tfl.cast"(%253) : (tensor<?x28x28x32xf32>) -> tensor<?x28x28x32xf16>
%255 = "tf.AddV2"(%237, %254) {device = ""} : (tensor<?x28x28x32xf16>, tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf16>
%256 = "tf.Conv2D"(%255, %112) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x32xf16>, tensor<1x1x32x192xf16>) -> tensor<?x28x28x192xf16>
%257 = "tfl.cast"(%256) : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf32
%258 = "tfl.mul"(%257, %113) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%259 = "tfl.add"(%258, %114) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%260 = "tfl.cast"(%259) : (tensor<?x28x28x192xf32>) -> tensor<?x28x28x192xf16
%261 = "tf.Relu6"(%260) {device = ""} : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf16>
%262 = "tf.DepthwiseConv2dNative"(%261, %109) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1]}> {device = ""} : (tensor<?x28x28x192xf16>, tensor<3x3x192x1xf16>) -> tensor<?x28x28x192xf16>
%263 = "tfl.cast"(%262) : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf32
%264 = "tfl.mul"(%263, %110) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%265 = "tfl.add"(%264, %111) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%266 = "tfl.cast"(%265) : (tensor<?x28x28x192xf32>) -> tensor<?x28x28x192xf16
%267 = "tf.Relu6"(%266) {device = ""} : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf16>
%268 = "tf.Conv2D"(%267, %115) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x192xf16>, tensor<1x1x192x32xf16>) -> tensor<?x28x28x32xf16>
%269 = "tfl.cast"(%268) : (tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf32>
%270 = "tfl.mul"(%269, %116) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x32xf32>, tensor<32xf32>) -> tensor<?x28x28x32xf32>
```

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%271 = "tfl.add"(%270, %117) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x32xf32>, tensor<32xf32>) -> tensor<?x28x28x32xf32>
%272 = "tfl.cast"(%271) : (tensor<?x28x28x32xf32>) -> tensor<?x28x28x32xf16>
%273 = "tf.AddV2"(%255, %272) {device = ""} : (tensor<?x28x28x32xf16>, tensor<?x28x28x32xf16>) -> tensor<?x28x28x32xf16>
%274 = "tf.Conv2D"(%273, %121) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x28x28x32xf16>, tensor<1x1x32x192xf16>) -> tensor<?x28x28x192xf16>
%275 = "tfl.cast"(%274) : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf32>
%276 = "tfl.mul"(%275, %122) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%277 = "tfl.add"(%276, %123) <{fused_activation_function = "NONE"}> : (tensor<?x28x28x192xf32>, tensor<192xf32>) -> tensor<?x28x28x192xf32>
%278 = "tfl.cast"(%277) : (tensor<?x28x28x192xf32>) -> tensor<?x28x28x192xf16>
%279 = "tf.Relu6"(%278) {device = ""} : (tensor<?x28x28x192xf16>) -> tensor<?x28x28x192xf16>
%280 = "tf.Pad"(%279, %5) {device = ""} : (tensor<?x28x28x192xf16>, tensor<4x2xi32>) -> tensor<?x29x29x192xf16>
%281 = "tf.DepthwiseConv2dNative"(%280, %118) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2, 1]}> {device = ""} : (tensor<?x29x29x192xf16>, tensor<3x3x192x1xf16>) -> tensor<?x14x14x192xf16>
%282 = "tfl.cast"(%281) : (tensor<?x14x14x192xf16>) -> tensor<?x14x14x192xf32>
%283 = "tfl.mul"(%282, %119) <{fused_activation_function = "NONE"}> : (tensor<?x14x14x192xf32>, tensor<192xf32>) -> tensor<?x14x14x192xf32>
%284 = "tfl.add"(%283, %120) <{fused_activation_function = "NONE"}> : (tensor<?x14x14x192xf32>, tensor<192xf32>) -> tensor<?x14x14x192xf32>
%285 = "tfl.cast"(%284) : (tensor<?x14x14x192xf32>) -> tensor<?x14x14x192xf16>
%286 = "tf.Relu6"(%285) {device = ""} : (tensor<?x14x14x192xf16>) -> tensor<?x14x14x192xf16>
%287 = "tf.Conv2D"(%286, %124) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x192xf16>, tensor<1x1x192x64xf16>) -> tensor<?x14x14x64xf16>
%288 = "tfl.cast"(%287) : (tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf32>
%289 = "tfl.mul"(%288, %125) <{fused_activation_function = "NONE"}> : (tensor<?x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%290 = "tfl.add"(%289, %126) <{fused_activation_function = "NONE"}> : (tensor<?x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%291 = "tfl.cast"(%290) : (tensor<?x14x14x64xf32>) -> tensor<?x14x14x64xf16>
%292 = "tf.Conv2D"(%291, %130) <{data_format = "NHWC", dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1], use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>, tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
%293 = "tfl.cast"(%292) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32>
%294 = "tfl.mul"(%293, %131) <{fused_activation_function = "NONE"}> : (tensor<?x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%295 = "tfl.add"(%294, %132) <{fused_activation_function = "NONE"}> : (tensor<?x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
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%296 = "tfl.cast"(%295) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%297 = "tf.Relu6"(%296) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%298 = "tf.DepthwiseConv2dNative"(%297, %127) <{data_format = "NHWC",␣
↪dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides =
↪[1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
%299 = "tfl.cast"(%298) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%300 = "tfl.mul"(%299, %128) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%301 = "tfl.add"(%300, %129) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%302 = "tfl.cast"(%301) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%303 = "tf.Relu6"(%302) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%304 = "tf.Conv2D"(%303, %133) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x64xf16>) -> tensor<?x14x14x64xf16>
%305 = "tfl.cast"(%304) : (tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf32>
%306 = "tfl.mul"(%305, %134) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%307 = "tfl.add"(%306, %135) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%308 = "tfl.cast"(%307) : (tensor<?x14x14x64xf32>) -> tensor<?x14x14x64xf16>
%309 = "tf.AddV2"(%291, %308) {device = ""} : (tensor<?x14x14x64xf16>, tensor ?
↪x14x14x64xf16>) -> tensor<?x14x14x64xf16>
%310 = "tf.Conv2D"(%309, %139) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
%311 = "tfl.cast"(%310) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%312 = "tfl.mul"(%311, %140) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%313 = "tfl.add"(%312, %141) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%314 = "tfl.cast"(%313) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%315 = "tf.Relu6"(%314) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%316 = "tf.DepthwiseConv2dNative"(%315, %136) <{data_format = "NHWC",␣
↪dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides =
↪[1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
%317 = "tfl.cast"(%316) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%318 = "tfl.mul"(%317, %137) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%319 = "tfl.add"(%318, %138) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%320 = "tfl.cast"(%319) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
```

```
%321 = "tf.Relu6"(%320) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%322 = "tf.Conv2D"(%321, %142) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x64xf16>) -> tensor<?x14x14x64xf16>
%323 = "tfl.cast"(%322) : (tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf32>
%324 = "tfl.mul"(%323, %143) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%325 = "tfl.add"(%324, %144) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%326 = "tfl.cast"(%325) : (tensor<?x14x14x64xf32>) -> tensor<?x14x14x64xf16>
%327 = "tf.AddV2"(%309, %326) {device = ""} : (tensor<?x14x14x64xf16>, tensor ?
↪x14x14x64xf16>) -> tensor<?x14x14x64xf16>
%328 = "tf.Conv2D"(%327, %148) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
%329 = "tfl.cast"(%328) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%330 = "tfl.mul"(%329, %149) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%331 = "tfl.add"(%330, %150) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%332 = "tfl.cast"(%331) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%333 = "tf.Relu6"(%332) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%334 = "tf.DepthwiseConv2dNative"(%333, %145) <{data_format = "NHWC",␣
↪dilations = [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides =
↪[1, 1, 1, 1]}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<3x3x384x1xf16>) -> tensor<?x14x14x384xf16>
%335 = "tfl.cast"(%334) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%336 = "tfl.mul"(%335, %146) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%337 = "tfl.add"(%336, %147) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%338 = "tfl.cast"(%337) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%339 = "tf.Relu6"(%338) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%340 = "tf.Conv2D"(%339, %151) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x64xf16>) -> tensor<?x14x14x64xf16>
%341 = "tfl.cast"(%340) : (tensor<?x14x14x64xf16>) -> tensor<?x14x14x64xf32>
%342 = "tfl.mul"(%341, %152) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%343 = "tfl.add"(%342, %153) <{fused_activation_function = "NONE"}> : (tensor ?
↪x14x14x64xf32>, tensor<64xf32>) -> tensor<?x14x14x64xf32>
%344 = "tfl.cast"(%343) : (tensor<?x14x14x64xf32>) -> tensor<?x14x14x64xf16>
%345 = "tf.AddV2"(%327, %344) {device = ""} : (tensor<?x14x14x64xf16>, tensor ?
↪x14x14x64xf16>) -> tensor<?x14x14x64xf16>
```

```
%346 = "tf.Conv2D"(%345, %13) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x64xf16>,␣
↪tensor<1x1x64x384xf16>) -> tensor<?x14x14x384xf16>
%347 = "tfl.cast"(%346) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%348 = "tfl.mul"(%347, %14) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%349 = "tfl.add"(%348, %15) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%350 = "tfl.cast"(%349) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%351 = "tf.Relu6"(%350) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%352 = "tf.DepthwiseConv2dNative"(%351, %10) <{data_format = "NHWC", dilation
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,
↪1]}> {device = ""} : (tensor<?x14x14x384xf16>, tensor<3x3x384x1xf16>) ->␣
↪tensor<?x14x14x384xf16>
%353 = "tfl.cast"(%352) : (tensor<?x14x14x384xf16>) -> tensor<?x14x14x384xf32
%354 = "tfl.mul"(%353, %11) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%355 = "tfl.add"(%354, %12) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x384xf32>, tensor<384xf32>) -> tensor<?x14x14x384xf32>
%356 = "tfl.cast"(%355) : (tensor<?x14x14x384xf32>) -> tensor<?x14x14x384xf16
%357 = "tf.Relu6"(%356) {device = ""} : (tensor<?x14x14x384xf16>) -> tensor<?
↪x14x14x384xf16>
%358 = "tf.Conv2D"(%357, %16) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x384xf16>,␣
↪tensor<1x1x384x96xf16>) -> tensor<?x14x14x96xf16>
%359 = "tfl.cast"(%358) : (tensor<?x14x14x96xf16>) -> tensor<?x14x14x96xf32>
%360 = "tfl.mul"(%359, %17) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x96xf32>, tensor<96xf32>) -> tensor<?x14x14x96xf32>
%361 = "tfl.add"(%360, %18) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x96xf32>, tensor<96xf32>) -> tensor<?x14x14x96xf32>
%362 = "tfl.cast"(%361) : (tensor<?x14x14x96xf32>) -> tensor<?x14x14x96xf16>
%363 = "tf.Conv2D"(%362, %22) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x96xf16>,␣
↪tensor<1x1x96x576xf16>) -> tensor<?x14x14x576xf16>
%364 = "tfl.cast"(%363) : (tensor<?x14x14x576xf16>) -> tensor<?x14x14x576xf32
%365 = "tfl.mul"(%364, %23) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%366 = "tfl.add"(%365, %24) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%367 = "tfl.cast"(%366) : (tensor<?x14x14x576xf32>) -> tensor<?x14x14x576xf16
%368 = "tf.Relu6"(%367) {device = ""} : (tensor<?x14x14x576xf16>) -> tensor<?
↪x14x14x576xf16>
%369 = "tf.DepthwiseConv2dNative"(%368, %19) <{data_format = "NHWC", dilation
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,
↪1]}> {device = ""} : (tensor<?x14x14x576xf16>, tensor<3x3x576x1xf16>) ->␣
↪tensor<?x14x14x576xf16>
```

```
%370 = "tfl.cast"(%369) : (tensor<?x14x14x576xf16>) -> tensor<?x14x14x576xf32
%371 = "tfl.mul"(%370, %20) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%372 = "tfl.add"(%371, %21) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%373 = "tfl.cast"(%372) : (tensor<?x14x14x576xf32>) -> tensor<?x14x14x576xf16
%374 = "tf.Relu6"(%373) {device = ""} : (tensor<?x14x14x576xf16>) -> tensor<?
↪x14x14x576xf16>
%375 = "tf.Conv2D"(%374, %25) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<1x1x576x96xf16>) -> tensor<?x14x14x96xf16>
%376 = "tfl.cast"(%375) : (tensor<?x14x14x96xf16>) -> tensor<?x14x14x96xf32>
%377 = "tfl.mul"(%376, %26) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x96xf32>, tensor<96xf32>) -> tensor<?x14x14x96xf32>
%378 = "tfl.add"(%377, %27) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x96xf32>, tensor<96xf32>) -> tensor<?x14x14x96xf32>
%379 = "tfl.cast"(%378) : (tensor<?x14x14x96xf32>) -> tensor<?x14x14x96xf16>
%380 = "tf.AddV2"(%362, %379) {device = ""} : (tensor<?x14x14x96xf16>, tensor ?
↪x14x14x96xf16>) -> tensor<?x14x14x96xf16>
%381 = "tf.Conv2D"(%380, %31) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x96xf16>,␣
↪tensor<1x1x96x576xf16>) -> tensor<?x14x14x576xf16>
%382 = "tfl.cast"(%381) : (tensor<?x14x14x576xf16>) -> tensor<?x14x14x576xf32
%383 = "tfl.mul"(%382, %32) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%384 = "tfl.add"(%383, %33) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%385 = "tfl.cast"(%384) : (tensor<?x14x14x576xf32>) -> tensor<?x14x14x576xf16
%386 = "tf.Relu6"(%385) {device = ""} : (tensor<?x14x14x576xf16>) -> tensor<?
↪x14x14x576xf16>
%387 = "tf.DepthwiseConv2dNative"(%386, %28) <{data_format = "NHWC", dilation
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,
↪1]}> {device = ""} : (tensor<?x14x14x576xf16>, tensor<3x3x576x1xf16>) ->␣
↪tensor<?x14x14x576xf16>
%388 = "tfl.cast"(%387) : (tensor<?x14x14x576xf16>) -> tensor<?x14x14x576xf32
%389 = "tfl.mul"(%388, %29) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%390 = "tfl.add"(%389, %30) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%391 = "tfl.cast"(%390) : (tensor<?x14x14x576xf32>) -> tensor<?x14x14x576xf16
%392 = "tf.Relu6"(%391) {device = ""} : (tensor<?x14x14x576xf16>) -> tensor<?
↪x14x14x576xf16>
%393 = "tf.Conv2D"(%392, %34) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<1x1x576x96xf16>) -> tensor<?x14x14x96xf16>
%394 = "tfl.cast"(%393) : (tensor<?x14x14x96xf16>) -> tensor<?x14x14x96xf32>
```

```
%395 = "tfl.mul"(%394, %35) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x96xf32>, tensor<96xf32>) -> tensor<?x14x14x96xf32>
%396 = "tfl.add"(%395, %36) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x96xf32>, tensor<96xf32>) -> tensor<?x14x14x96xf32>
%397 = "tfl.cast"(%396) : (tensor<?x14x14x96xf32>) -> tensor<?x14x14x96xf16>
%398 = "tf.AddV2"(%380, %397) {device = ""} : (tensor<?x14x14x96xf16>, tensor ?
↪x14x14x96xf16>) -> tensor<?x14x14x96xf16>
%399 = "tf.Conv2D"(%398, %40) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x14x14x96xf16>,␣
↪tensor<1x1x96x576xf16>) -> tensor<?x14x14x576xf16>
%400 = "tfl.cast"(%399) : (tensor<?x14x14x576xf16>) -> tensor<?x14x14x576xf32
%401 = "tfl.mul"(%400, %41) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%402 = "tfl.add"(%401, %42) <{fused_activation_function = "NONE"}> : (tensor<
↪x14x14x576xf32>, tensor<576xf32>) -> tensor<?x14x14x576xf32>
%403 = "tfl.cast"(%402) : (tensor<?x14x14x576xf32>) -> tensor<?x14x14x576xf16
%404 = "tf.Relu6"(%403) {device = ""} : (tensor<?x14x14x576xf16>) -> tensor<?
↪x14x14x576xf16>
%405 = "tf.Pad"(%404, %5) {device = ""} : (tensor<?x14x14x576xf16>,␣
↪tensor<4x2xi32>) -> tensor<?x15x15x576xf16>
%406 = "tf.DepthwiseConv2dNative"(%405, %37) <{data_format = "NHWC", dilation
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "VALID", strides = [1, 2, 2 ␣
↪1]}> {device = ""} : (tensor<?x15x15x576xf16>, tensor<3x3x576x1xf16>) ->␣
↪tensor<?x7x7x576xf16>
%407 = "tfl.cast"(%406) : (tensor<?x7x7x576xf16>) -> tensor<?x7x7x576xf32>
%408 = "tfl.mul"(%407, %38) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x576xf32>, tensor<576xf32>) -> tensor<?x7x7x576xf32>
%409 = "tfl.add"(%408, %39) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x576xf32>, tensor<576xf32>) -> tensor<?x7x7x576xf32>
%410 = "tfl.cast"(%409) : (tensor<?x7x7x576xf32>) -> tensor<?x7x7x576xf16>
%411 = "tf.Relu6"(%410) {device = ""} : (tensor<?x7x7x576xf16>) -> tensor<?
↪x7x7x576xf16>
%412 = "tf.Conv2D"(%411, %43) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x576xf16>,␣
↪tensor<1x1x576x160xf16>) -> tensor<?x7x7x160xf16>
%413 = "tfl.cast"(%412) : (tensor<?x7x7x160xf16>) -> tensor<?x7x7x160xf32>
%414 = "tfl.mul"(%413, %44) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x160xf32>, tensor<160xf32>) -> tensor<?x7x7x160xf32>
%415 = "tfl.add"(%414, %45) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x160xf32>, tensor<160xf32>) -> tensor<?x7x7x160xf32>
%416 = "tfl.cast"(%415) : (tensor<?x7x7x160xf32>) -> tensor<?x7x7x160xf16>
%417 = "tf.Conv2D"(%416, %49) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x160xf16>,␣
↪tensor<1x1x160x960xf16>) -> tensor<?x7x7x960xf16>
%418 = "tfl.cast"(%417) : (tensor<?x7x7x960xf16>) -> tensor<?x7x7x960xf32>
%419 = "tfl.mul"(%418, %50) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
```

```
%420 = "tfl.add"(%419, %51) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%421 = "tfl.cast"(%420) : (tensor<?x7x7x960xf32>) -> tensor<?x7x7x960xf16>
%422 = "tf.Relu6"(%421) {device = ""} : (tensor<?x7x7x960xf16>) -> tensor<?
↪x7x7x960xf16>
%423 = "tf.DepthwiseConv2dNative"(%422, %46) <{data_format = "NHWC", dilation␣
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1]}> {device = ""} : (tensor<?x7x7x960xf16>, tensor<3x3x960x1xf16>) -> tensor<?
↪x7x7x960xf16>
%424 = "tfl.cast"(%423) : (tensor<?x7x7x960xf16>) -> tensor<?x7x7x960xf32>
%425 = "tfl.mul"(%424, %47) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%426 = "tfl.add"(%425, %48) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%427 = "tfl.cast"(%426) : (tensor<?x7x7x960xf32>) -> tensor<?x7x7x960xf16>
%428 = "tf.Relu6"(%427) {device = ""} : (tensor<?x7x7x960xf16>) -> tensor<?
↪x7x7x960xf16>
%429 = "tf.Conv2D"(%428, %52) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<1x1x960x160xf16>) -> tensor<?x7x7x160xf16>
%430 = "tfl.cast"(%429) : (tensor<?x7x7x160xf16>) -> tensor<?x7x7x160xf32>
%431 = "tfl.mul"(%430, %53) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x160xf32>, tensor<160xf32>) -> tensor<?x7x7x160xf32>
%432 = "tfl.add"(%431, %54) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x160xf32>, tensor<160xf32>) -> tensor<?x7x7x160xf32>
%433 = "tfl.cast"(%432) : (tensor<?x7x7x160xf32>) -> tensor<?x7x7x160xf16>
%434 = "tf.AddV2"(%416, %433) {device = ""} : (tensor<?x7x7x160xf16>, tensor<
↪x7x7x160xf16>) -> tensor<?x7x7x160xf16>
%435 = "tf.Conv2D"(%434, %58) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x160xf16>,␣
↪tensor<1x1x160x960xf16>) -> tensor<?x7x7x960xf16>
%436 = "tfl.cast"(%435) : (tensor<?x7x7x960xf16>) -> tensor<?x7x7x960xf32>
%437 = "tfl.mul"(%436, %59) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%438 = "tfl.add"(%437, %60) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%439 = "tfl.cast"(%438) : (tensor<?x7x7x960xf32>) -> tensor<?x7x7x960xf16>
%440 = "tf.Relu6"(%439) {device = ""} : (tensor<?x7x7x960xf16>) -> tensor<?
↪x7x7x960xf16>
%441 = "tf.DepthwiseConv2dNative"(%440, %55) <{data_format = "NHWC", dilation␣
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,␣
↪1]}> {device = ""} : (tensor<?x7x7x960xf16>, tensor<3x3x960x1xf16>) -> tensor<?
↪x7x7x960xf16>
%442 = "tfl.cast"(%441) : (tensor<?x7x7x960xf16>) -> tensor<?x7x7x960xf32>
%443 = "tfl.mul"(%442, %56) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%444 = "tfl.add"(%443, %57) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
```

```
%445 = "tfl.cast"(%444) : (tensor<?x7x7x960xf32>) -> tensor<?x7x7x960xf16>
%446 = "tf.Relu6"(%445) {device = ""} : (tensor<?x7x7x960xf16>) -> tensor<?
↪x7x7x960xf16>
%447 = "tf.Conv2D"(%446, %61) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<1x1x960x160xf16>) -> tensor<?x7x7x160xf16>
%448 = "tfl.cast"(%447) : (tensor<?x7x7x160xf16>) -> tensor<?x7x7x160xf32>
%449 = "tfl.mul"(%448, %62) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x160xf32>, tensor<160xf32>) -> tensor<?x7x7x160xf32>
%450 = "tfl.add"(%449, %63) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x160xf32>, tensor<160xf32>) -> tensor<?x7x7x160xf32>
%451 = "tfl.cast"(%450) : (tensor<?x7x7x160xf32>) -> tensor<?x7x7x160xf16>
%452 = "tf.AddV2"(%434, %451) {device = ""} : (tensor<?x7x7x160xf16>, tensor<
↪x7x7x160xf16>) -> tensor<?x7x7x160xf16>
%453 = "tf.Conv2D"(%452, %67) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x160xf16>,␣
↪tensor<1x1x160x960xf16>) -> tensor<?x7x7x960xf16>
%454 = "tfl.cast"(%453) : (tensor<?x7x7x960xf16>) -> tensor<?x7x7x960xf32>
%455 = "tfl.mul"(%454, %68) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%456 = "tfl.add"(%455, %69) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%457 = "tfl.cast"(%456) : (tensor<?x7x7x960xf32>) -> tensor<?x7x7x960xf16>
%458 = "tf.Relu6"(%457) {device = ""} : (tensor<?x7x7x960xf16>) -> tensor<?
↪x7x7x960xf16>
%459 = "tf.DepthwiseConv2dNative"(%458, %64) <{data_format = "NHWC", dilation
↪= [1, 1, 1, 1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1,
↪1]}> {device = ""} : (tensor<?x7x7x960xf16>, tensor<3x3x960x1xf16>) -> tensor
↪x7x7x960xf16>
%460 = "tfl.cast"(%459) : (tensor<?x7x7x960xf16>) -> tensor<?x7x7x960xf32>
%461 = "tfl.mul"(%460, %65) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%462 = "tfl.add"(%461, %66) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x960xf32>, tensor<960xf32>) -> tensor<?x7x7x960xf32>
%463 = "tfl.cast"(%462) : (tensor<?x7x7x960xf32>) -> tensor<?x7x7x960xf16>
%464 = "tf.Relu6"(%463) {device = ""} : (tensor<?x7x7x960xf16>) -> tensor<?
↪x7x7x960xf16>
%465 = "tf.Conv2D"(%464, %70) <{data_format = "NHWC", dilations = [1, 1, 1,␣
↪1], explicit_paddings = [], padding = "SAME", strides = [1, 1, 1, 1],␣
↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x960xf16>,␣
↪tensor<1x1x960x320xf16>) -> tensor<?x7x7x320xf16>
%466 = "tfl.cast"(%465) : (tensor<?x7x7x320xf16>) -> tensor<?x7x7x320xf32>
%467 = "tfl.mul"(%466, %71) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x320xf32>, tensor<320xf32>) -> tensor<?x7x7x320xf32>
%468 = "tfl.add"(%467, %72) <{fused_activation_function = "NONE"}> : (tensor<
↪x7x7x320xf32>, tensor<320xf32>) -> tensor<?x7x7x320xf32>
%469 = "tfl.cast"(%468) : (tensor<?x7x7x320xf32>) -> tensor<?x7x7x320xf16>
```

```
  %470 = "tf.Conv2D"(%469, %157) <{data_format = "NHWC", dilations = [1, 1, 1,␣
 ↪1], explicit_paddings = [], padding = "VALID", strides = [1, 1, 1, 1],␣
 ↪use_cudnn_on_gpu = true}> {device = ""} : (tensor<?x7x7x320xf16>,␣
 ↪tensor<1x1x320x1280xf16>) -> tensor<?x7x7x1280xf16>
  %471 = "tfl.cast"(%470) : (tensor<?x7x7x1280xf16>) -> tensor<?x7x7x1280xf32>
  %472 = "tfl.mul"(%471, %158) <{fused_activation_function = "NONE"}> : (tensor<?
 ↪x7x7x1280xf32>, tensor<1280xf32>) -> tensor<?x7x7x1280xf32>
  %473 = "tfl.add"(%472, %159) <{fused_activation_function = "NONE"}> : (tensor<?
 ↪x7x7x1280xf32>, tensor<1280xf32>) -> tensor<?x7x7x1280xf32>
  %474 = "tfl.cast"(%473) : (tensor<?x7x7x1280xf32>) -> tensor<?x7x7x1280xf16>
  %475 = "tf.Relu6"(%474) {device = ""} : (tensor<?x7x7x1280xf16>) -> tensor<?
 ↪x7x7x1280xf16>
  %476 = "tfl.cast"(%475) : (tensor<?x7x7x1280xf16>) -> tensor<?x7x7x1280xf32>
  %477 = "tfl.mean"(%476, %4) <{keep_dims = false}> : (tensor<?x7x7x1280xf32>,␣
 ↪tensor<2xi32>) -> tensor<?x1280xf32>
  %478 = "tfl.cast"(%477) : (tensor<?x1280xf32>) -> tensor<?x1280xf16>
  %479 = "tfl.cast"(%478) : (tensor<?x1280xf16>) -> tensor<?x1280xf32>
  %480 = "tfl.mul"(%479, %6) <{fused_activation_function = "NONE"}> : (tensor<?
 ↪x1280xf32>, tensor<1280xf32>) -> tensor<?x1280xf32>
  %481 = "tfl.add"(%480, %7) <{fused_activation_function = "NONE"}> : (tensor<?
 ↪x1280xf32>, tensor<1280xf32>) -> tensor<?x1280xf32>
  %482 = "tfl.cast"(%481) : (tensor<?x1280xf32>) -> tensor<?x1280xf16>
  %483 = "tf.MatMul"(%482, %2) <{grad_a = false, grad_b = false, transpose_a =␣
 ↪false, transpose_b = true}> : (tensor<?x1280xf16>, tensor<128x1280xf16>) ->␣
 ↪tensor<?x128xf16>
  %484 = "tf.BiasAdd"(%483, %9) <{data_format = "NHWC"}> {device = ""} :␣
 ↪(tensor<?x128xf16>, tensor<128xf16>) -> tensor<?x128xf16>
  %485 = "tf.Relu"(%484) {device = ""} : (tensor<?x128xf16>) -> tensor<?x128xf1 >
  %486 = "tf.MatMul"(%485, %1) <{grad_a = false, grad_b = false, transpose_a =␣
 ↪false, transpose_b = true}> : (tensor<?x128xf16>, tensor<64x128xf16>) ->␣
 ↪tensor<?x64xf16>
  %487 = "tf.BiasAdd"(%486, %8) <{data_format = "NHWC"}> {device = ""} :␣
 ↪(tensor<?x64xf16>, tensor<64xf16>) -> tensor<?x64xf16>
  %488 = "tf.Relu"(%487) {device = ""} : (tensor<?x64xf16>) -> tensor<?x64xf16>
  %489 = "tfl.cast"(%488) : (tensor<?x64xf16>) -> tensor<?x64xf32>
  %490 = "tfl.fully_connected"(%489, %0, %3) <{fused_activation_function =␣
 ↪"NONE", keep_num_dims = false, weights_format = "DEFAULT"}> : (tensor<?
 ↪x64xf32>, tensor<4x64xf32>, tensor<4xf32>) -> tensor<?x4xf32>
  %491 = "tfl.softmax"(%490) <{beta = 1.000000e+00 : f32}> : (tensor<?x4xf32>)␣
 ↪-> tensor<?x4xf32>
  "func.return"(%491) : (tensor<?x4xf32>) -> ()
}) {tf.entry_function = {control_outputs = "", inputs =␣
 ↪"serving_default_input_layer_1:0", outputs = "StatefulPartitionedCall_1:0"},␣
 ↪tf_saved_model.exported_names = ["serving_default"]} : () -> ()
```

[ ]:
```
best_model = tf.keras.models.load_model(
    'best_rose_model.keras',
```

```
    compile=False
)

converter = tf.lite.TFLiteConverter.from_keras_model(best_model)

converter.target_spec.supported_ops = [
    tf.lite.OpsSet.TFLITE_BUILTINS,
    tf.lite.OpsSet.SELECT_TF_OPS
]

converter.optimizations = []

tflite_model = converter.convert()

with open('rose_disease_model.tflite', 'wb') as f:
    f.write(tflite_model)

print("TFLite model saved successfully.")
```

1.18 17. TFLite Verification

Load the Float32 TFLite model using `tf.lite.Interpreter`, run inference on the test set, and compare accuracy with the original Keras model to verify that no accuracy degradation occurred during conversion.

[ ]:
```
#   Keras baseline accuracy (already evaluated in Section 9)
keras_acc = best_model.evaluate(test_ds, verbose=0)[1]
print(f'Keras model test accuracy : {keras_acc:.4f}')

#   Float32 TFLite inference on the test set
interpreter = tf.lite.Interpreter(model_path='rose_disease_model.tflite')
interpreter.allocate_tensors()

input_details  = interpreter.get_input_details()
output_details = interpreter.get_output_details()

tflite_preds = []
for path, true_label in zip(x_test_paths, y_test):
    img = tf.keras.utils.load_img(path, target_size=(IMG_SIZE, IMG_SIZE))
    img = tf.keras.utils.img_to_array(img)
    img = tf.keras.applications.mobilenet_v2.preprocess_input(img)
    img = np.expand_dims(img, axis=0).astype(np.float32)
    interpreter.set_tensor(input_details[0]['index'], img)
    interpreter.invoke()
    output = interpreter.get_tensor(output_details[0]['index'])
    tflite_preds.append(np.argmax(output[0]))
```

```
tflite_preds = np.array(tflite_preds)
tflite_acc   = np.mean(tflite_preds == y_test)
print(f'TFLite (float32) test accuracy : {tflite_acc:.4f}')
print(f'Accuracy delta (Keras - TFLite): {keras_acc - tflite_acc:.4f}')

if abs(keras_acc - tflite_acc) < 0.005:
    print('  TFLite conversion verified - no significant accuracy drift.')
else:
    print('  Consider representative dataset calibration or quantisation-aware␣
 ↪training.')
```

1.19 18. Bias-Reduction Recommendations

- Keep **class weights** as the only imbalance strategy unless a full ablation study supports another method.
- Prioritize collecting more diverse yellow-leaf samples (lighting, camera, age/stage) to reduce color shortcut learning.
- Preserve color-aware augmentation (brightness/contrast/saturation/hue) to improve robustness to yellow hue shifts.
- Track per-class precision/recall each run; optimize for balanced recall rather than overall accuracy only.
- If yellow over-prediction persists, calibrate probabilities (temperature scaling) before deployment.